



BITS Pilani

Embedded System Design

Sangmeshwar Kendre



AEZG512/AELZG512/DMZG512
Embedded System Design
Remote Lab Contact Session-01

Contents



- Instructions
- Remote Lab Demonstration

Instructions to be followed

- You MUST have BITS ID and password to access the lab. Use same credentials as that of <http://elearn.bits-pilani.ac.in>
- Everyone should complete the class recordings before accessing the remote lab.
- Unfortunately, no additional slots will be available for you under any circumstances.
- Ensure the recordings, documents, and remote assignments to be performed are in the elearn portal or under the teams before accessing the lab.
- Only one slot will be available with two hours of duration.
- The cancellation of slots will only be possible before 24 hours, and you can rebook again, which is only possible for one time.

Instructions to be followed



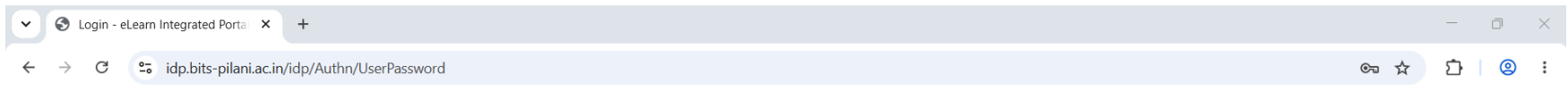
- Verify that RDP is enabled on your computer; if not, follow the steps: When you're ready, select Start > System > Remote Desktop, and turn on Enable Remote Desktop
- Don't use office laptops as they are equipped with security firewalls.

Remote Lab Booking



Go to <https://prayogshala.bits-pilani.ac.in/auth/login> Click on Continue with BITS SSO

Enter your BITS ID and password to login (Use same credentials as that of <http://elearn.bits-pilani.ac.in>). Click on Login



**WORK
INTEGRATED
LEARNING
PROGRAMMES**

BITS Email Address

kendres.shankarrao@wilp.bits-pilani.ac.in

Password

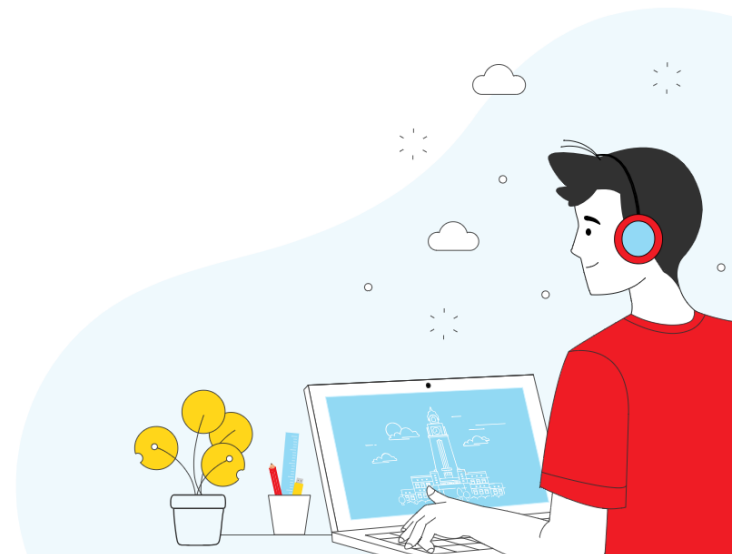
.....

☐ Remember Me

[Forgot password?](#)

Log In >

[Click here for login instructions](#)



After successful login

Browser: BITS Lab Management Portal | URL: prayogshala.bits-pilani.ac.in/dashboard



BITS Pilani
Work Integrated Learning Programmes

PRAYOGSHALA
LAB MANAGEMENT PORTAL

Faculty | 2 | Logout

Assigned Courses [View All Courses →](#)

AEZG512 REMOTE_DESKTOP

 **EMBEDDED SYSTEM DESIGN**

SEM 1/2026-27

Sem: 25 Jul, 2026 - 31 Dec, 2026

Lab Starts: 17 Aug, 2026

152 students enrolled

 View Activities

Assigned Labs

AEZG512 Booking Required Remote

 **EMBEDDED SYSTEM DESIGN**

 Book a Slot

 **KENDRE SANGMESH...**
FACULTY

For Remote Lab Booking



Select the course: EMBEDDED SYSTEM DESIGN and click on Book a slot

The screenshot displays the BITS Pilani Prayogshala Lab Management Portal. The browser address bar shows the URL `prayogshala.bits-pilani.ac.in/dashboard`. The page header includes the BITS Pilani logo, the portal name, a faculty dropdown menu, a notification bell with a red '2', and a 'Logout' link. The left sidebar contains a 'Dashboard' button and a 'My Courses' link. The main content area is titled 'Assigned Courses' and features a card for the 'EMBEDDED SYSTEM DESIGN' course (AEZG512, REMOTE_DESKTOP). The course details include the semester (SEM 1/2026-27), dates (25 Jul, 2026 - 31 Dec, 2026), lab start date (17 Aug, 2026), and 152 enrolled students. A 'View Activities' button is present. Below this, the 'Assigned Labs' section shows a card for the same course with a 'Booking Required' status and a 'Remote' icon. A 'Book a Slot' button is highlighted with a red rectangle. The bottom left corner shows the user's name 'KENDRE SANGMESH...' and 'FACULTY'.

Select the Lab: **ESD Lab**, Machine: **ESD1**, Date: **17-08-2026** and Time slot: **8:00 AM to 10:00 AM**. Click on **Confirm Booking**

BITS Lab Management Portal x +

← → ↺ 📄 prayogshala.bits-pilani.ac.in/dashboard ☆ 📄 👤 ⋮



PRAYOGSHALA
LAB MANAGEMENT PORTAL

AEZG512 REMOTE_DESKTOP

EMBEDDED SYSTEM DESIGN

SEM 12026-27

Term: 26 Jul, 2026 – 31 Dec, 2026

Lab Starts: 17 Aug, 2026

152 students enrolled

View Activities

Assigned Labs

AEZG512 Booking Requested Remote

EMBEDDED SYSTEM DESIGN

Book a Slot

KENDRE SANDHESH...
FACULTY

Book a Lab Slot

Select lab, machine, date and time slot

EMBEDDED SYSTEM DESIGN
AEZG512 Remote Desktop

Total Sessions **1** Remaining **1** Completed **0**

Lab & Machine

Lab * **ESD Lab** Machine * **ESD 1**

INSTALLED TOOLS **S32**

Date & Time

Select Date * **17-08-2026** Time Slot * **8:00 AM – 10:00 AM**

Monday, 17 Aug 2026

Cancel **Confirm Booking**

You will see booked lab session. Start Session button will be enabled at the appointed date and time only. Click on it

The screenshot shows the BITS Pilani Prayogshala Lab Management Portal. The top navigation bar includes the BITS Pilani logo and the text 'WORK INTEGRATED LEARNING PROGRAMMES'. The main header displays 'PRAYOGSHALA LAB MANAGEMENT PORTAL' and a 'Faculty' dropdown menu. The left sidebar contains 'Dashboard' and 'My Courses' links. The main content area is divided into sections: 'Assigned Labs' and 'Upcoming Lab Sessions'. Under 'Assigned Labs', there is a card for 'EMBEDDED SYSTEM DESIGN' (AEZG512) with a 'Booking Required' status and a 'Remote' icon. Below this, the 'Upcoming Lab Sessions' section lists a session for '17 Aug, 2026 Monday' from '8:00 AM – 10:00 AM' for 'EMBEDDED SYSTEM DESIGN' (AEZG512). This session is highlighted with a red border and shows a 'Confirmed' status and a 'Start Session' button.

Assigned Labs

AEZG512 Booking Required Remote

EMBEDDED SYSTEM DESIGN

Book a Slot

Upcoming Lab Sessions

17 Aug, 2026 Monday
8:00 AM – 10:00 AM

EMBEDDED SYSTEM DESIGN
AEZG512

Confirmed Start Session Cancel

To access remote lab



Click on **Start Session**

The screenshot shows the BITS Pilani Prayogshala Lab Management Portal. The browser address bar displays `prayogshala.bits-pilani.ac.in/dashboard`. The page header includes the BITS Pilani logo and the text "WORK INTEGRATED LEARNING PROGRAMMES". The main content area is titled "PRAYOGSHALA LAB MANAGEMENT PORTAL" and shows a summary of the current semester (Sem: 25 Jul, 2026 - 31 Dec, 2026) with 152 students enrolled. The "Assigned Labs" section features a card for "EMBEDDED SYSTEM DESIGN" (AEZG512) with a "Book a Slot" button. The "Upcoming Lab Sessions" section shows a session for "EMBEDDED SYSTEM DESIGN" (AEZG512) on 17 Aug, 2026, from 8:00 AM to 10:00 AM. A red box highlights the "Start Session" button in the upcoming sessions section.

Option-1: To connect remote through Web RDP PC, Click on **Connect Web RDP**

The screenshot displays the 'Remote Lab Session' interface for 'EMBEDDED SYSTEM DESIGN - AEZG512'. The interface includes a sidebar with navigation options like 'Dashboard' and 'My Courses', and a main content area with session statistics and connection credentials.

Remote Lab Session
EMBEDDED SYSTEM DESIGN - AEZG512

Session Statistics:

- TOTAL SESSIONS: 6
- BOOKED: 3
- AVAILABLE: 3

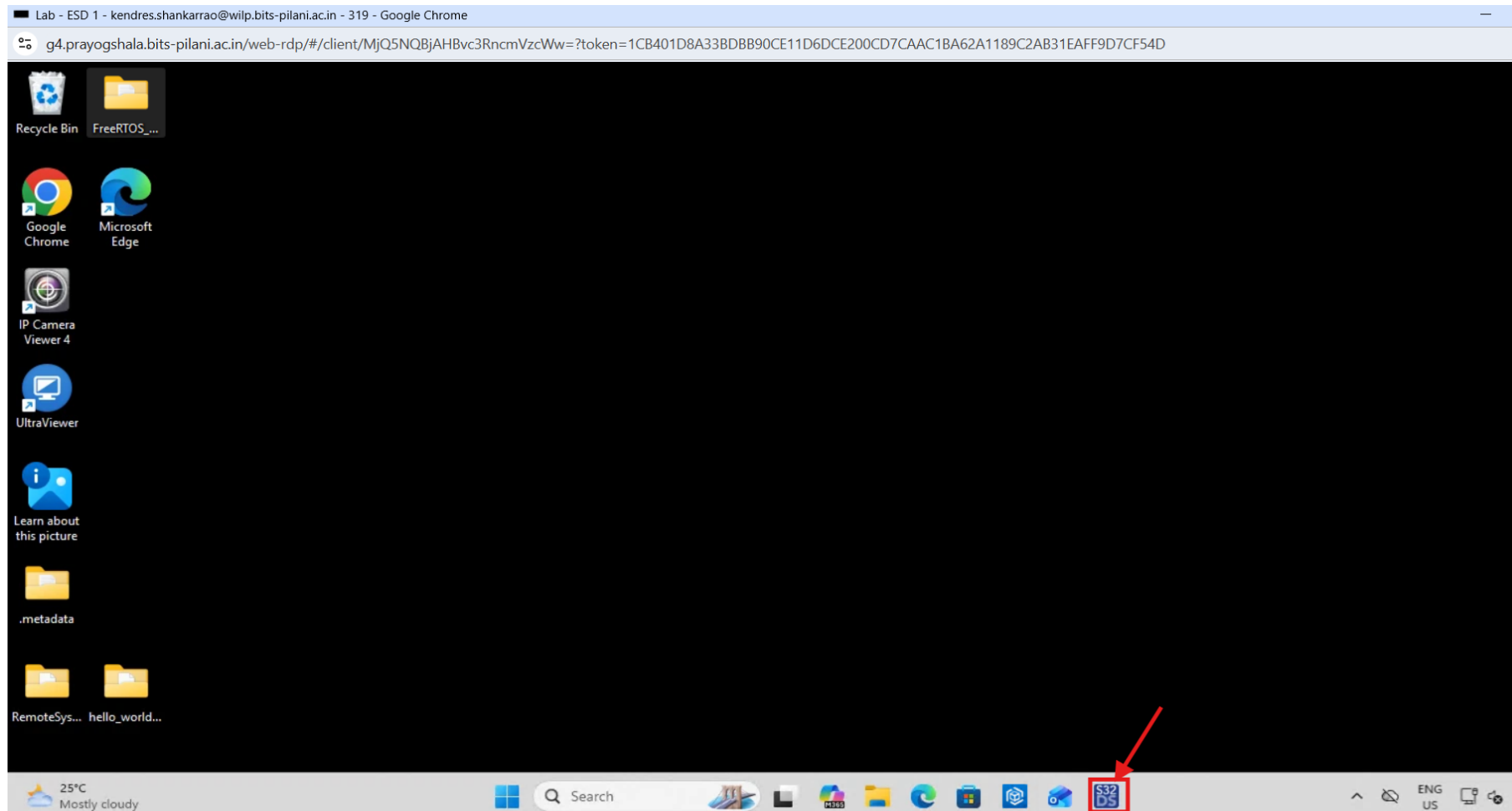
CONNECTION CREDENTIALS

Username: Argo-137
Password: Qg9G14dP

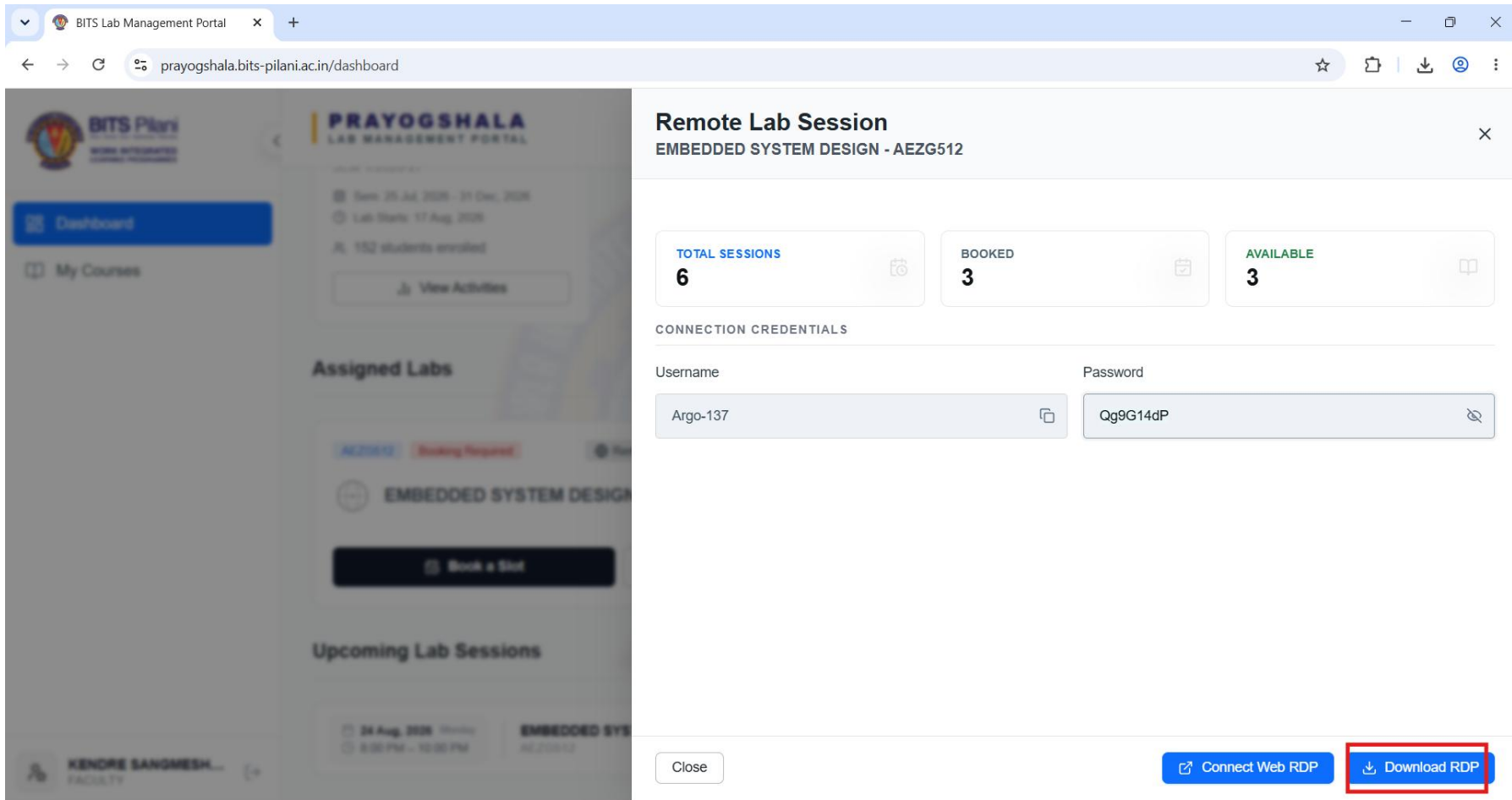
Buttons:

- Close
- Connect Web RDP (highlighted with a red box)
- Download RDP

Then it will be connected to remote lab using browser.



Option-2: Download RDP. Click on **Download RDP**



The screenshot displays the BITS Pilani Prayogshala Lab Management Portal. The main interface is titled "Remote Lab Session" for "EMBEDDED SYSTEM DESIGN - AEZG512". It shows session statistics: 6 total sessions, 3 booked, and 3 available. Below this, connection credentials are provided: Username "Argo-137" and Password "Qg9G14dP". At the bottom, there are two buttons: "Connect Web RDP" and "Download RDP", with the latter being highlighted by a red box.

Remote Lab Session
EMBEDDED SYSTEM DESIGN - AEZG512

TOTAL SESSIONS
6

BOOKED
3

AVAILABLE
3

CONNECTION CREDENTIALS

Username: Argo-137

Password: Qg9G14dP

[Close](#) [Connect Web RDP](#) [Download RDP](#)

Double Click on downloaded RDP file. Click on Connect

The screenshot shows the BITS Lab Management Portal interface. The browser address bar displays `prayogshala.bits-pilani.ac.in/dashboard`. The portal header includes the BITS Pilani logo and the text "PRAYOGSHALA LAB MANAGEMENT PORTAL". The main content area is titled "Remote Lab Session EMBEDDED SYSTEM DESIGN - AEZG512". Below this, there are sections for "Assigned Labs" and "Upcoming Lab Sessions". A security warning dialog is overlaid on the screen, titled "Remote Desktop Connection security warning". The dialog contains a "Caution: Unknown remote connection" message and a table with the following information:

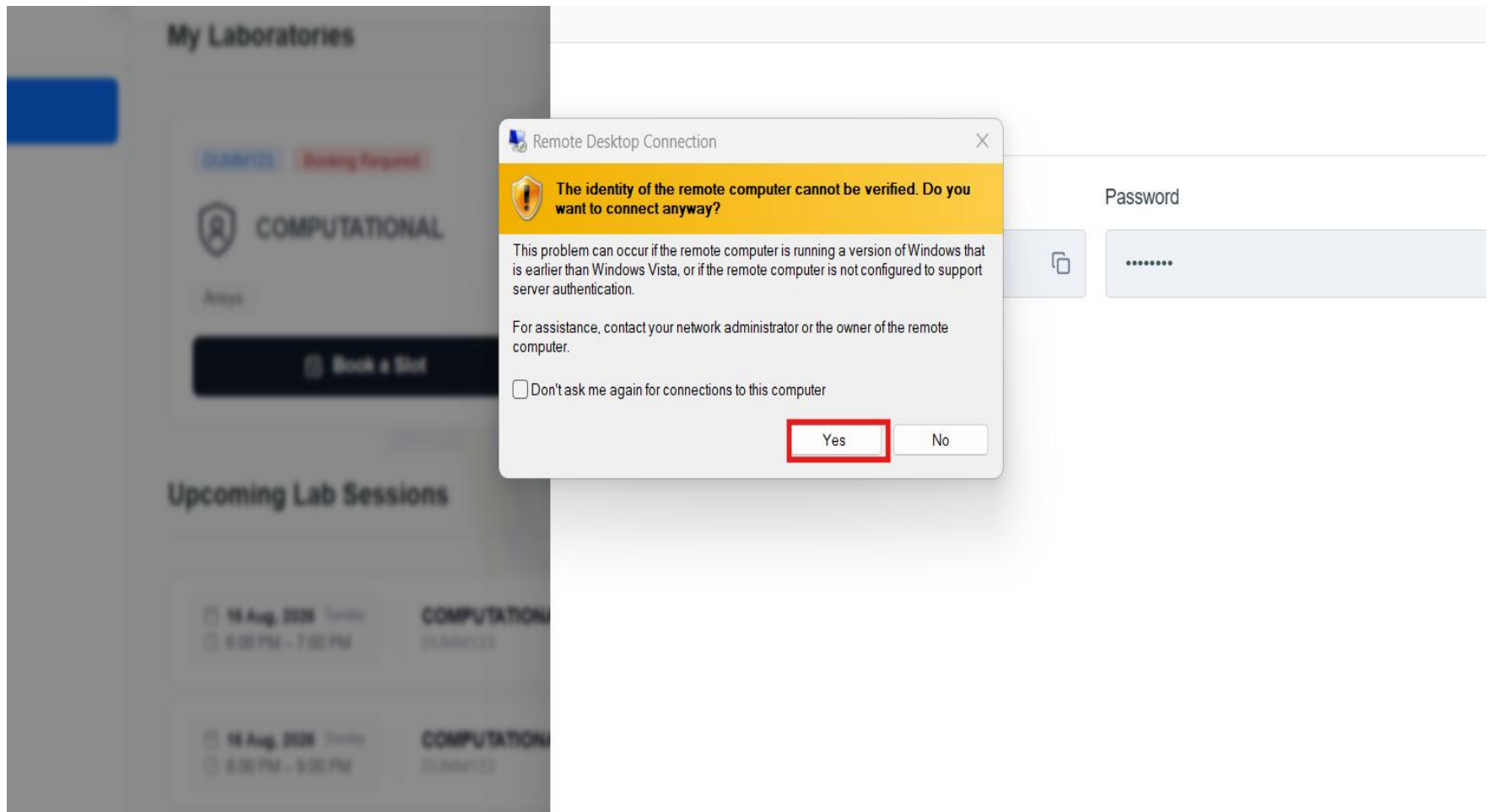
Property	Value
Publisher:	Unknown publisher
Type:	Remote Desktop Connection
Remote computer:	125.22.54.200

Below the table, there are checkboxes for allowing the remote computer to access resources on the local computer:

- ☐ Smart cards or Windows Hello for Business
- ☐ WebAuthn (passkeys and security keys)
- ☐ Clipboard
- ☐ Printers

At the bottom of the dialog, there are "Connect" and "Cancel" buttons. The "Connect" button is highlighted with a red box. The background portal interface also shows a "Password" field with the value "Qg9G14dP" and buttons for "Connect Web RDP" and "Download RDP".

Click Yes



Enter your Credentials given by the portal, password. Then click OK

The screenshot shows the BITS Lab Management Portal interface. A 'Remote Lab Session' window is open for 'EMBEDDED SYSTEM DESIGN - AEZG512'. It displays session statistics: 6 total sessions, 3 booked, and 3 available. Below this, the 'CONNECTION CREDENTIALS' section shows a 'Username' field with 'Argo-137' and a 'Password' field with masked characters. A Windows Security dialog box is overlaid on the portal, titled 'Enter your credentials'. It states: 'These credentials will be used to connect to 125.22.54.200.' The dialog shows the same 'Username' and 'Password' fields. Red arrows point from the portal's credential fields to the dialog's fields. The dialog has an 'OK' button and a 'Cancel' button. At the bottom of the portal, there are buttons for 'Connect Web RDP' and 'Download RDP'.

Windows Security

Enter your credentials

These credentials will be used to connect to 125.22.54.200.

Username: Argo-137

Password: [Masked]

☐ Remember me

More choices

OK Cancel

Remote Lab Session
EMBEDDED SYSTEM DESIGN - AEZG512

TOTAL SESSIONS: 6

BOOKED: 3

AVAILABLE: 3

CONNECTION CREDENTIALS

Username: Argo-137

Password: [Masked]

Close

Connect Web RDP Download RDP

Click on Yes.

BITS Lab Management Portal x +

prayogshala.bits-pilani.ac.in/dashboard

Remote Lab Session

EMBEDDED SYSTEM DESIGN - AEZG512

Remote Desktop Connection

The identity of the remote computer cannot be verified. Do you want to connect anyway?

The remote computer could not be authenticated due to problems with its security certificate. It may be unsafe to proceed.

Certificate name

Name in the certificate from the remote computer:
DESKTOP-DHEGII9

Certificate errors

The following errors were encountered while validating the remote computer's certificate:

The certificate is not from a trusted certifying authority.

Do you want to connect despite these certificate errors?

☐ Don't ask me again for connections to this computer

View certificate... **Yes** No

Close

Connect Web RDP Download RDP

Assigned Labs

EMBEDDED SYSTEM DESIGN - AEZG512

Booking Required

Upcoming Lab Sessions

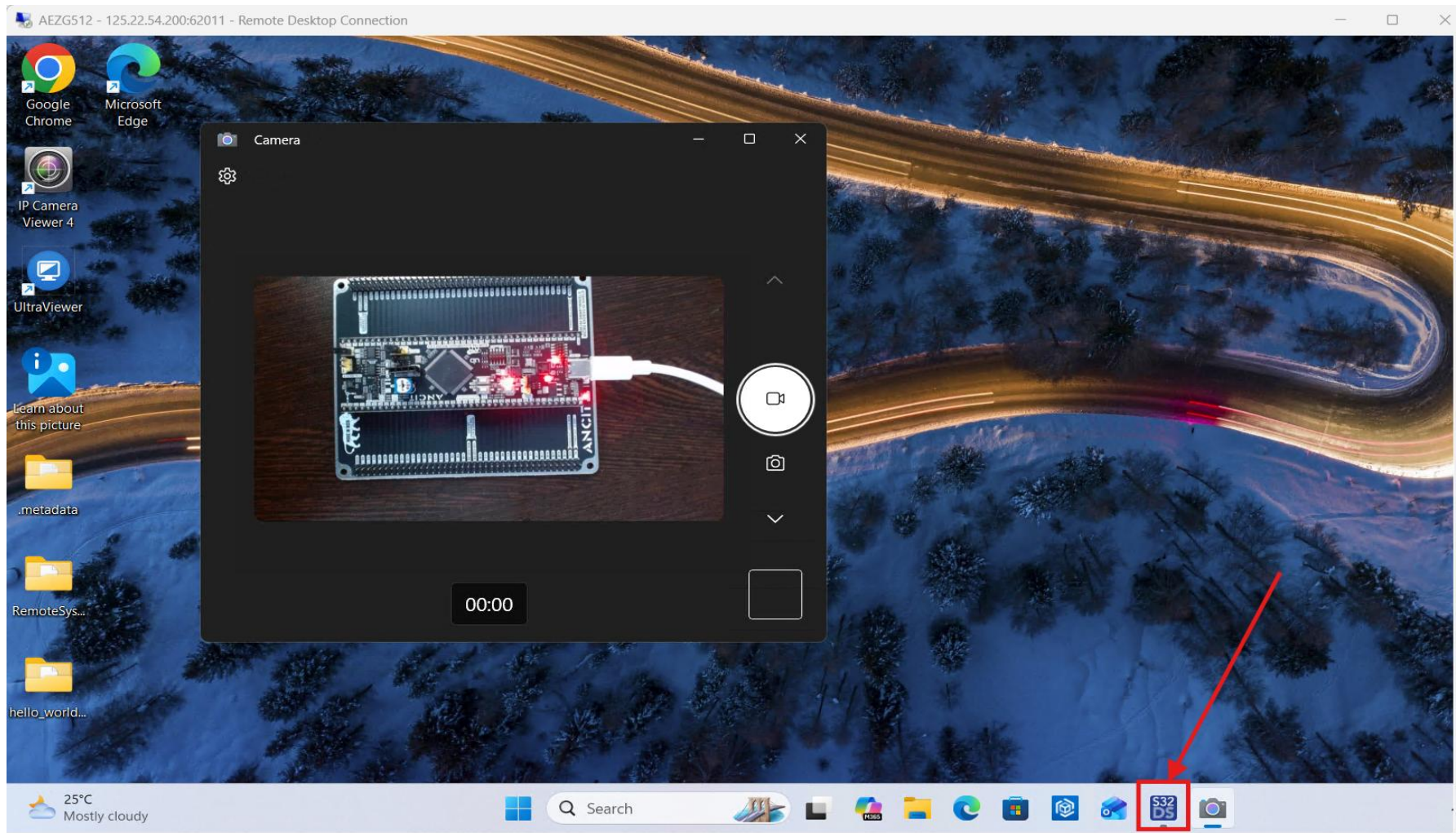
24 Aug 2026 Monday

8:00 PM - 10:00 PM

EMBEDDED SYSTEM DESIGN - AEZG512

KENDRE SANGMESH... FACULTY

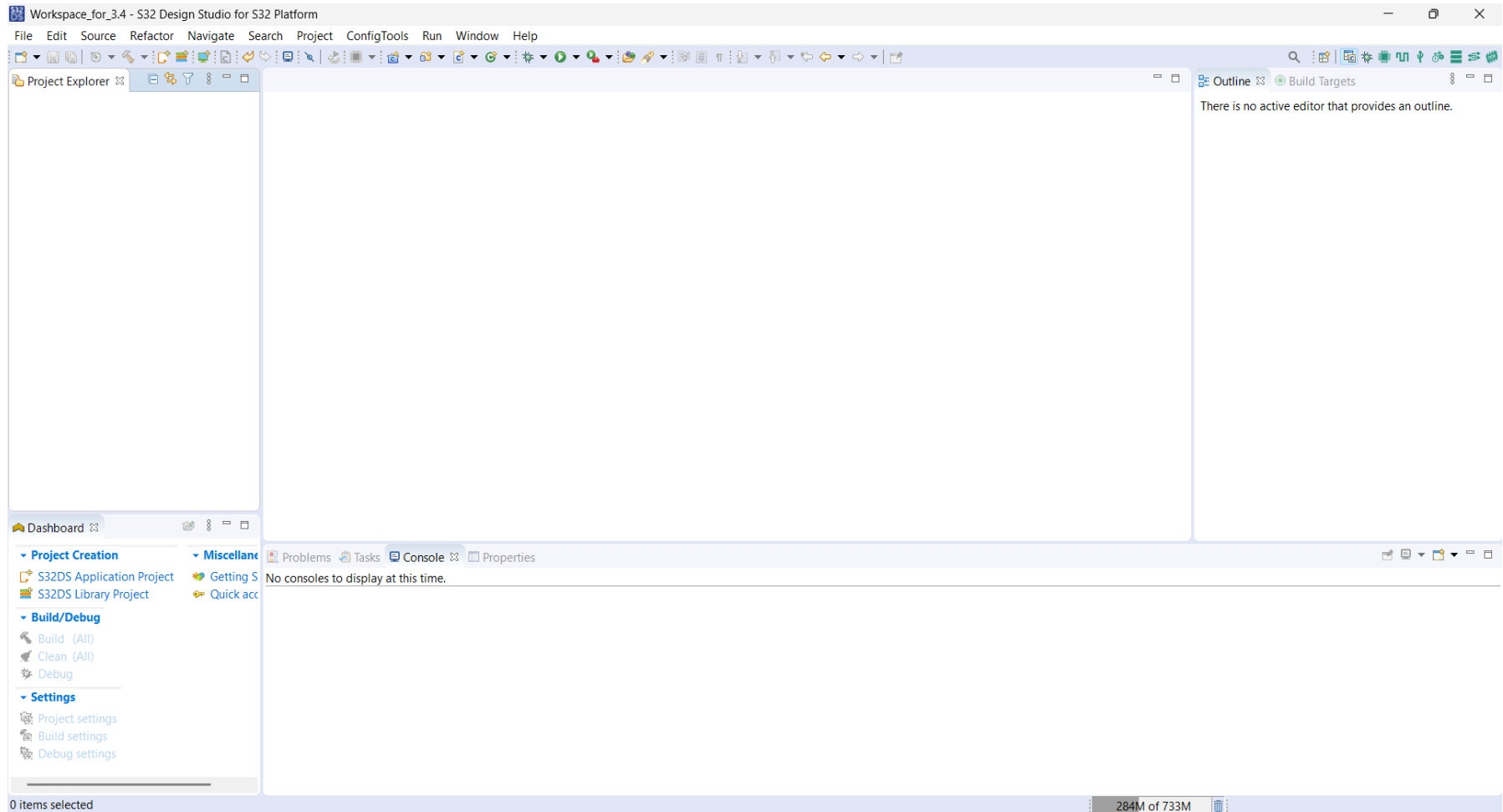
Remote Desktop opens. Start S32 Design studio and Camera from application menu.



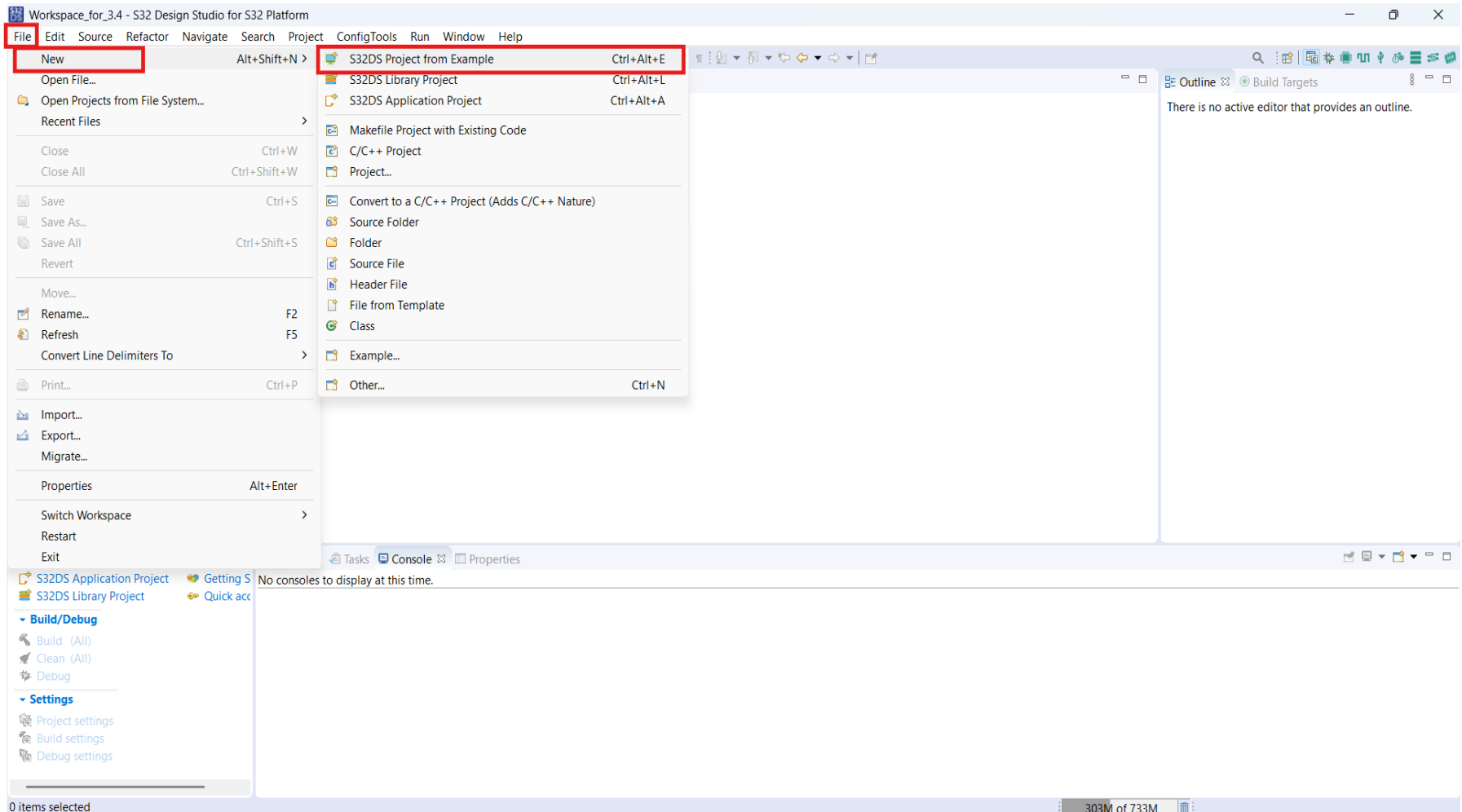


Project Creation in S32 Design Studio

S32 Design Studio for S32 Platform 3.4



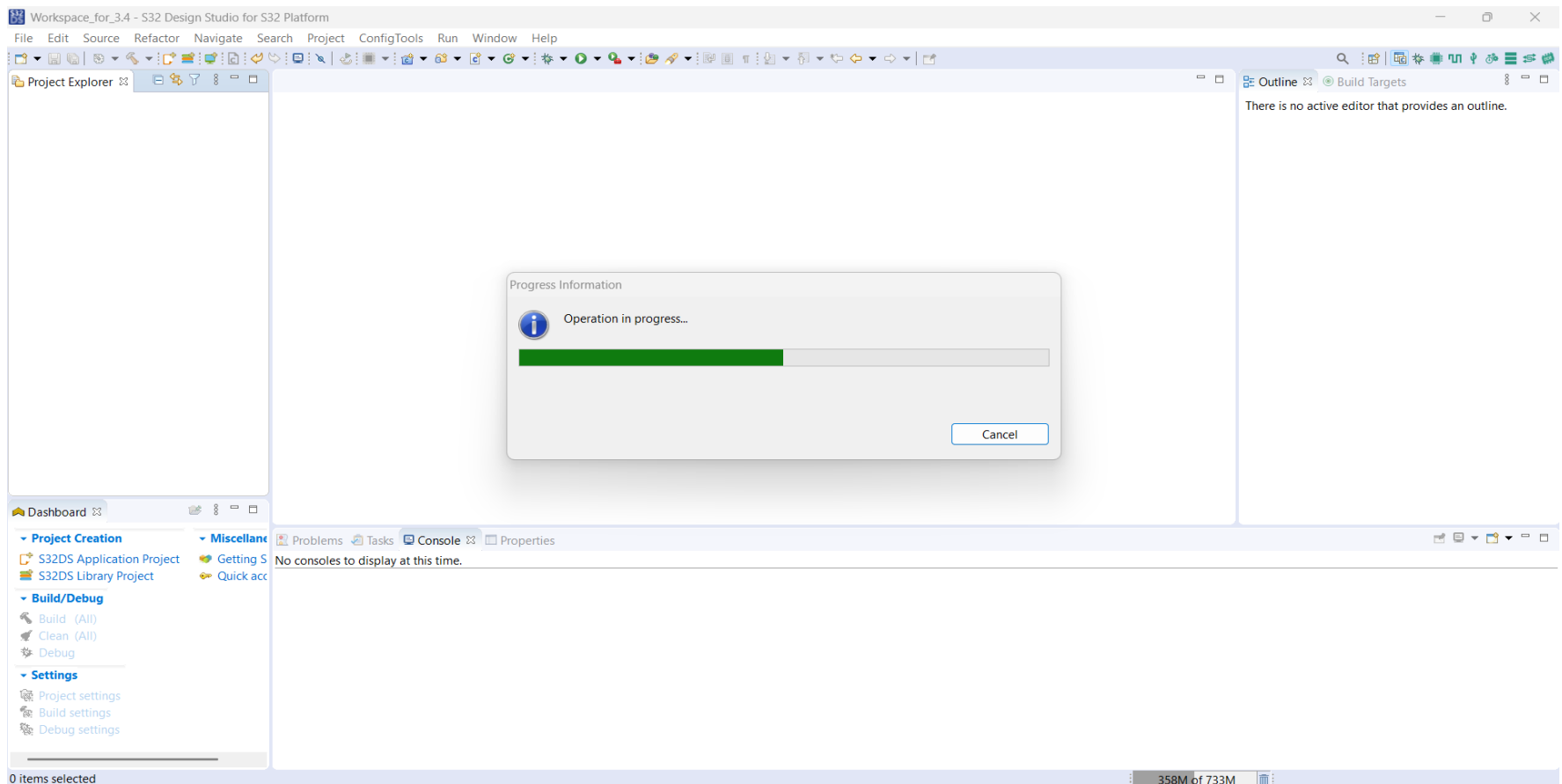
Navigate to File→New→S32DS Project from Example



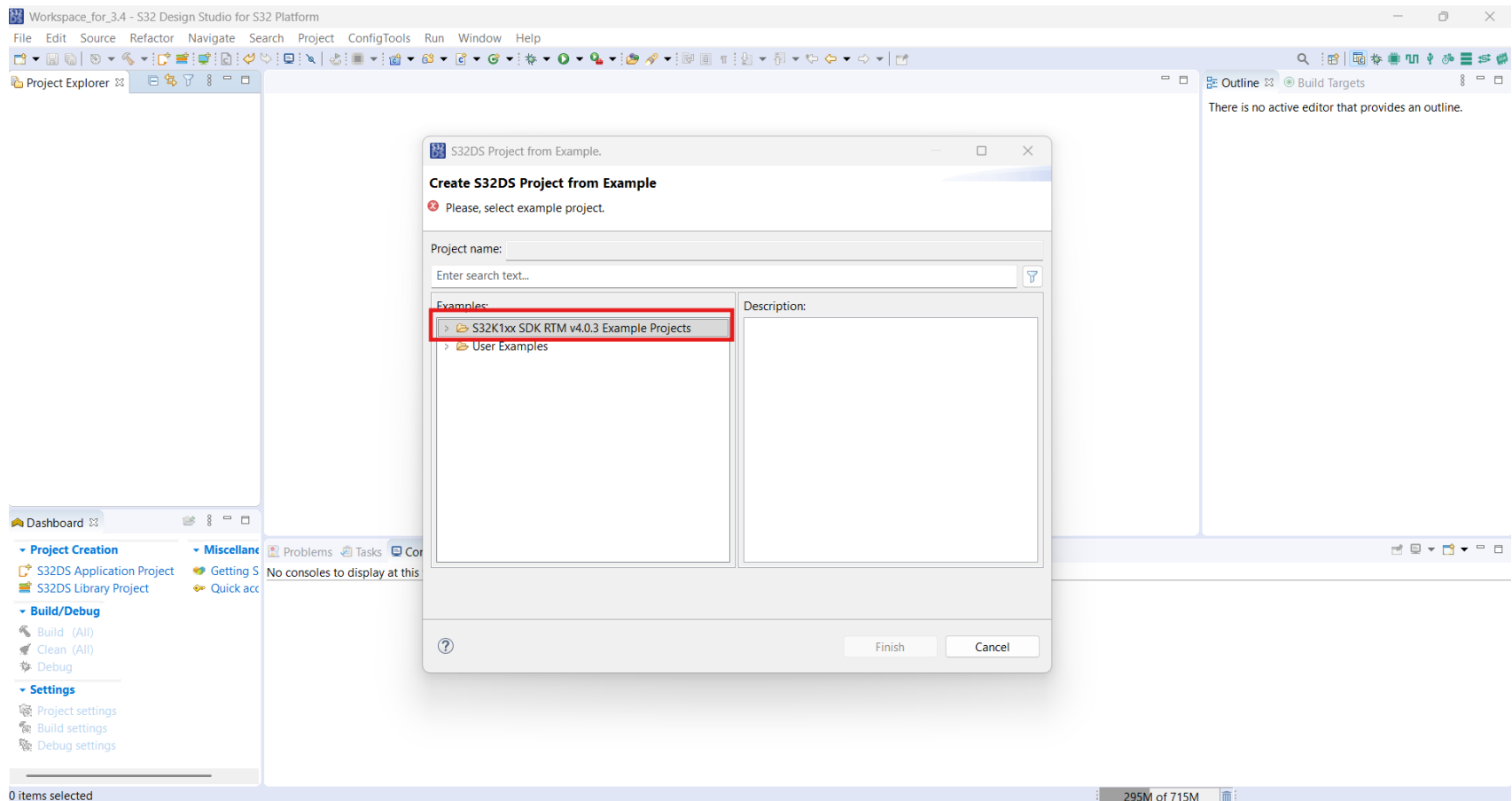
The screenshot shows the S32 Design Studio for S32 Platform IDE. The 'File' menu is open, and the 'New' option is selected. The 'New' submenu is also open, showing various project and file creation options. The 'S32DS Project from Example' option is highlighted, with the keyboard shortcut 'Ctrl+Alt+E' displayed next to it.

The main workspace area is empty, showing the 'Outline' and 'Build Targets' tabs. The status bar at the bottom indicates '0 items selected' and '303M of 733M'.

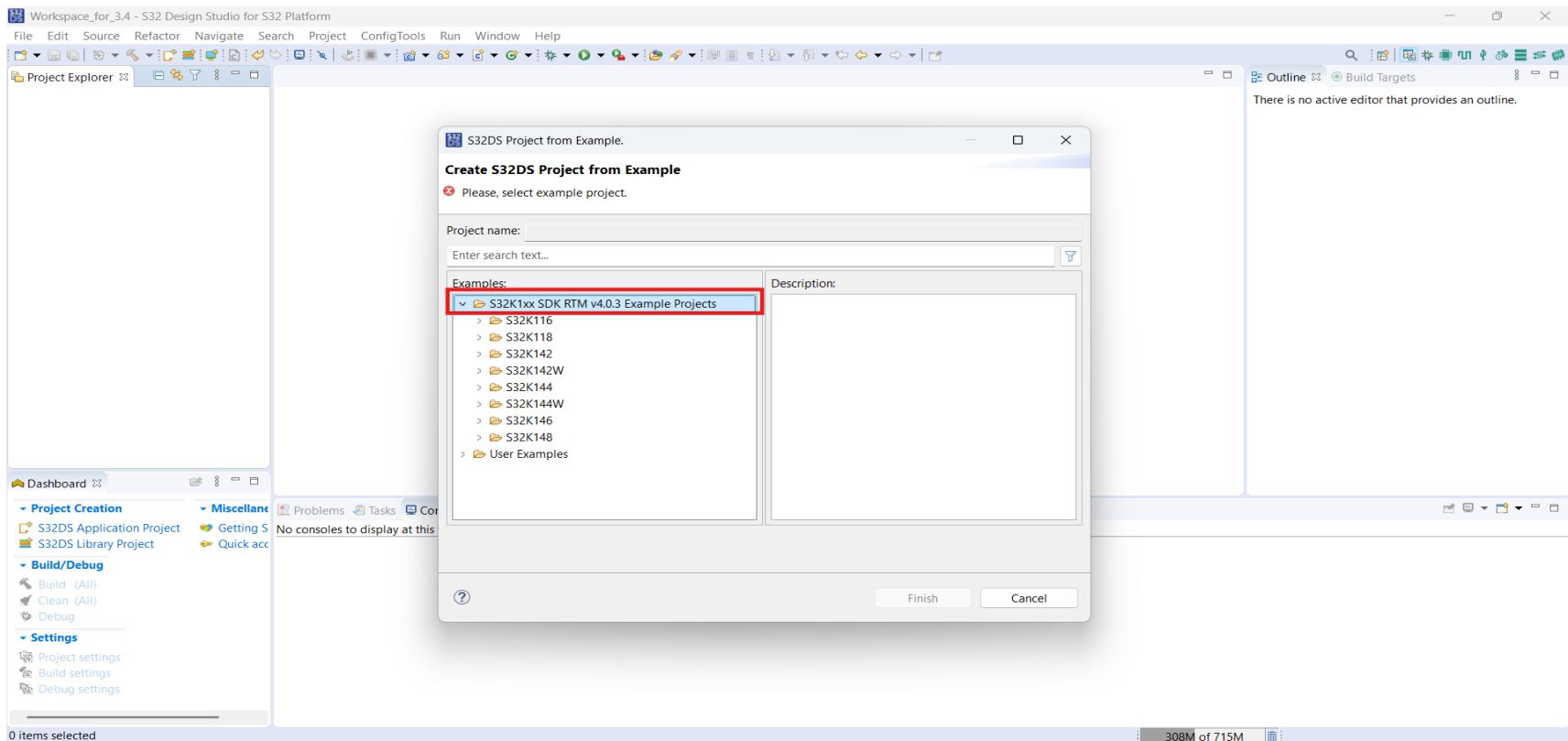
After that a window will pop-up like this, and it will be loading all demo examples from internal SDK that we have installed



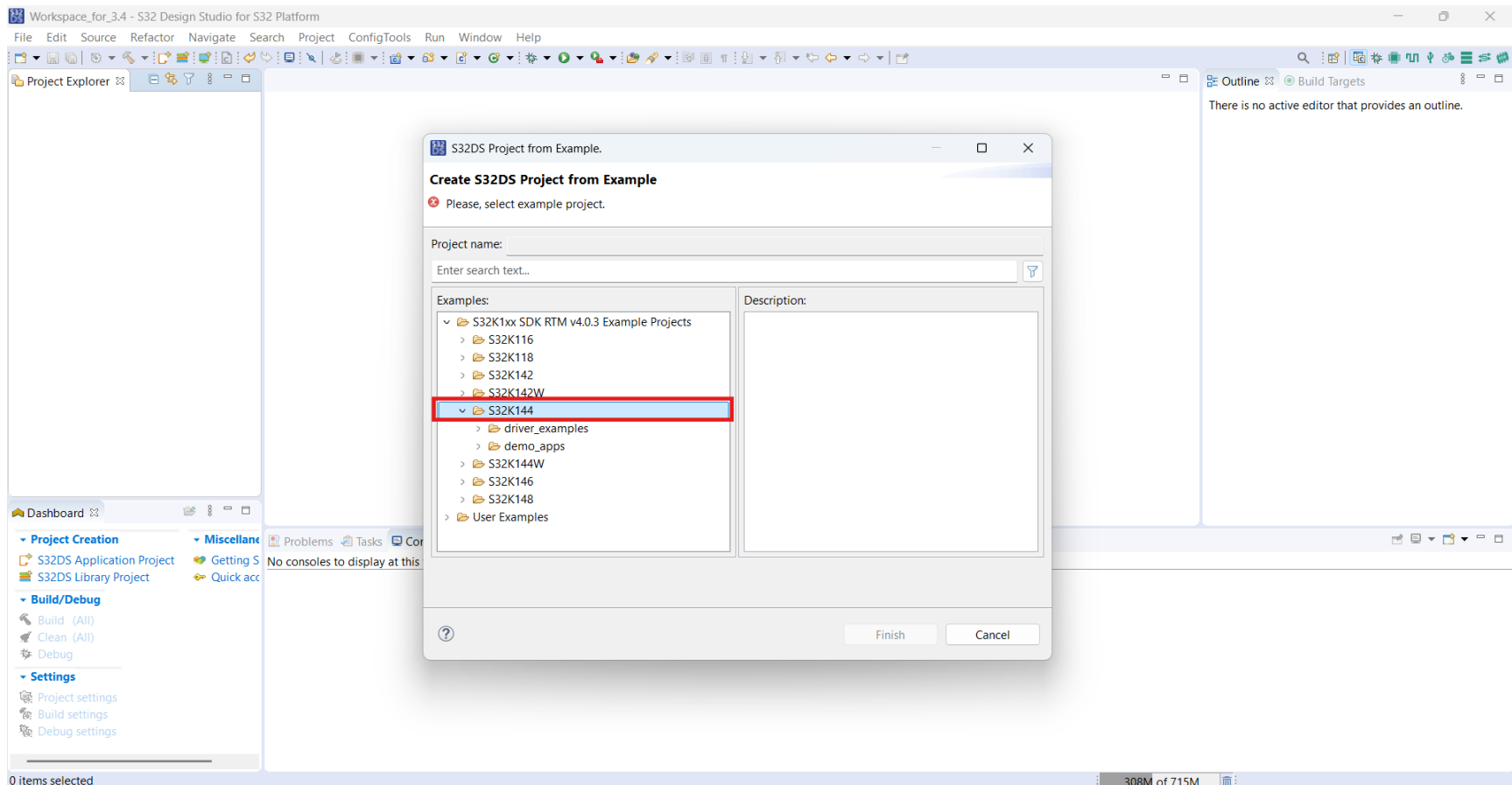
After successful loading, you will see S32K1xx SDK RTM v4.0.3 Example Projects



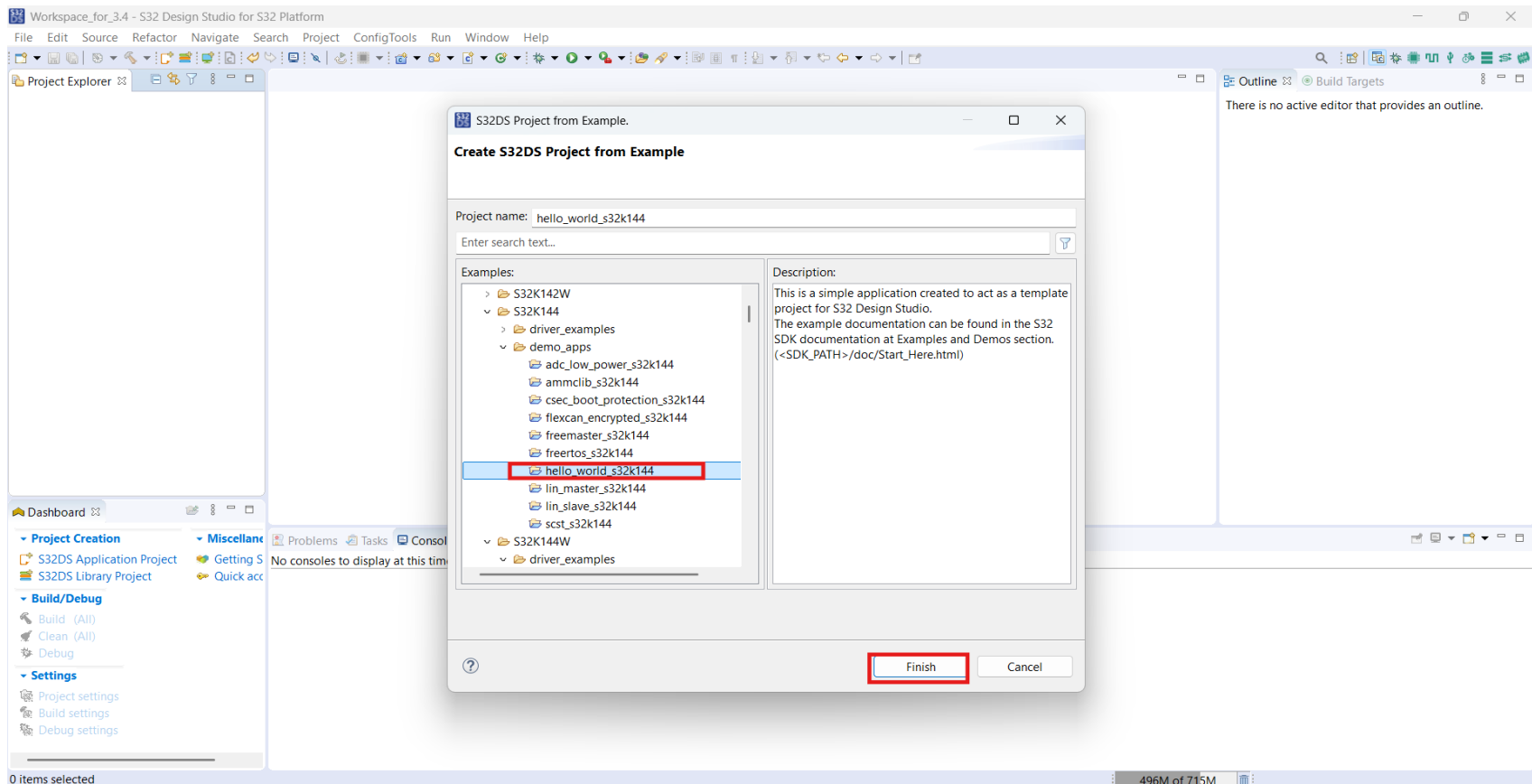
Now click on the SDK name: S32K1xx SDK RTM v4.0.3 Example Projects and we can see list of demo codes for all the MCU's of S32K1 family: S32K116, S32K118, S32K142, S32K142W, S32K144, S32K144W, S32K146, S32K148, as shown below.



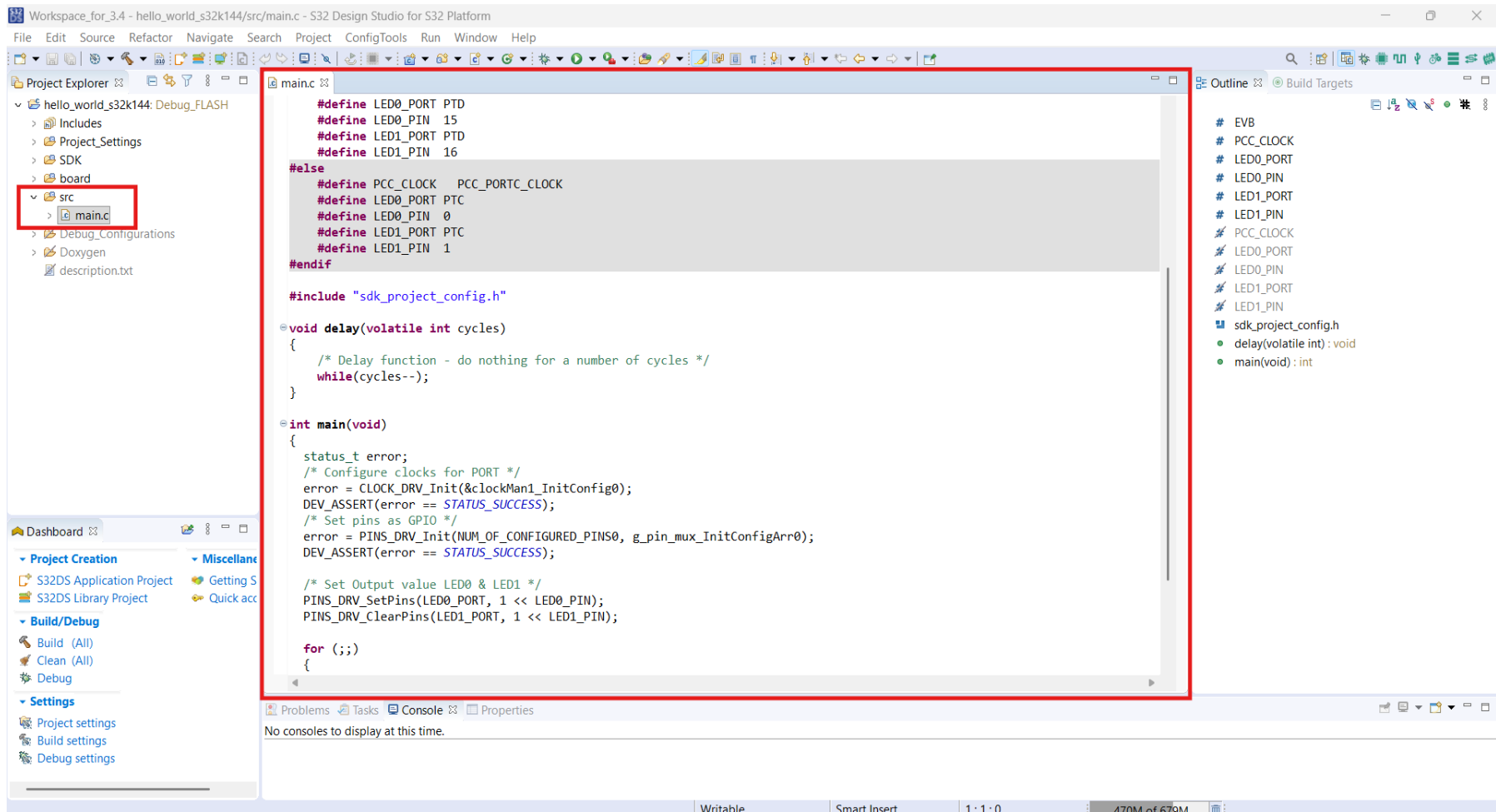
We are using development board which is made on S32K144 MCU, so we will be exploring demo examples of S32K144 MCU.



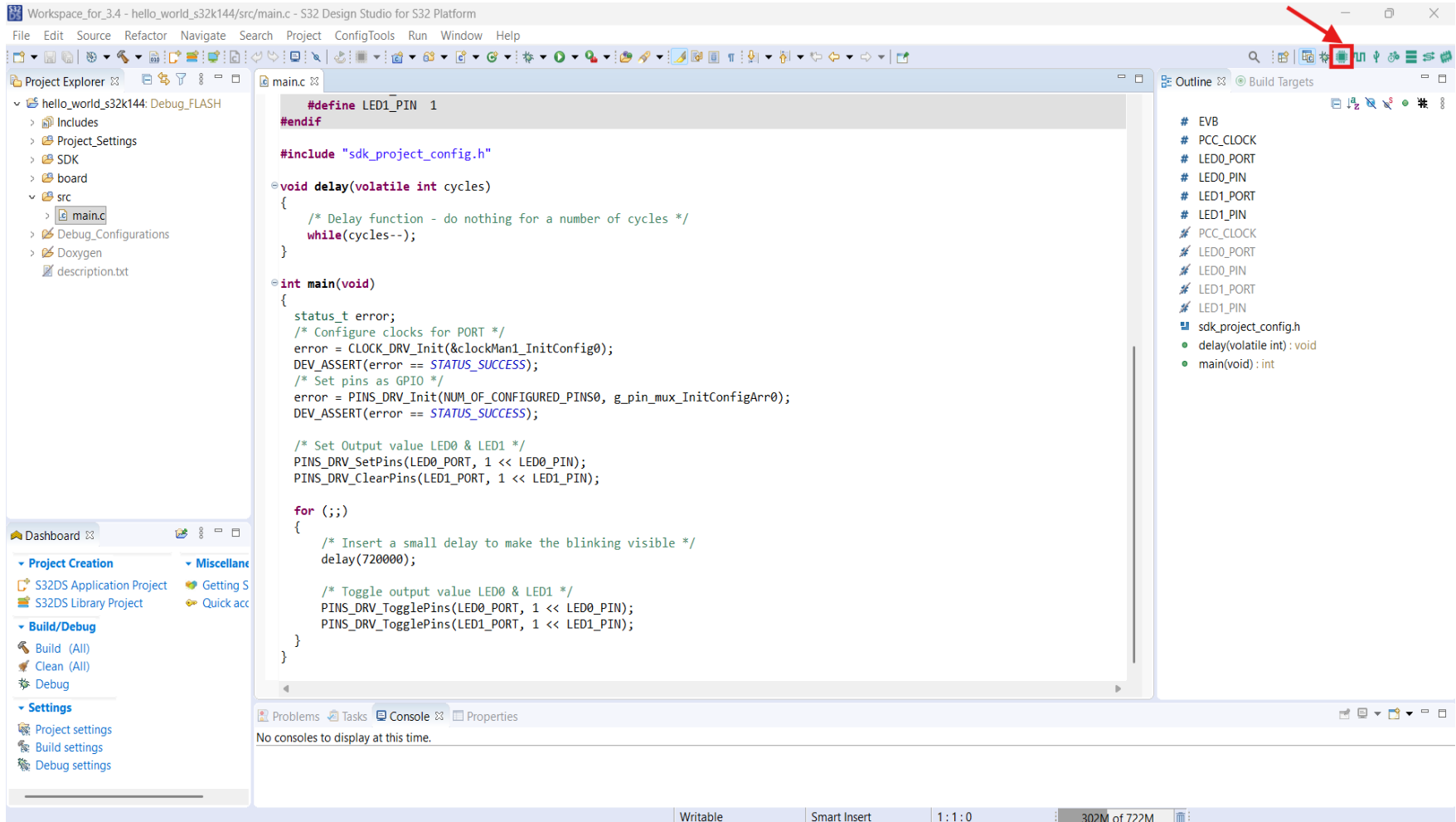
Select `hello_world_s32k144` example and click on Finish. This Project blinks the on board RGB LED which is attached at pin PD15 and PD16 of the S32K144 MCU



After opening the project and main.c file, window will look like this:



For Pin Routing Details: Click on Pins



Pinout table

SI	PIN	Label	Identifier	Description
1	PTD15	RGB_LED_RED	RGB_LED_RED	RGB Red LED, Active Low
2	PTD16	RGB_LED_GREEN	RGB_LED_GREEN	RGB Green LED, Active Low
3	PTD0	RGB_LED_BLUE	RGB_LED_BLUE	RGB Blue LED, Active Low
4	PTC12	USER_SW1	DIG_IP1	User Switch 1, On-board Pulled-up, Active Low
5	PTC13	USER_SW2	DIG_IP2	User Switch 2, On-board Pulled-up, Active Low
6	PTC14	ADC0_CH12	ANG_CH1	Potentiometer 10K ohms, Output Voltage: 0 to 3.3V
7	PTC7	LPUART1_TX	LPUART1_TX	UART Instance-1, Tx
8	PTC6	LPUART1_RX	LPUART1_RX	UART Instance-1, Rx
9	PTA2	LPI2C0_SDA	LPI2C0_SDA	I2C Instance-0, Data pin
10	PTA3	LPI2C0_SCL	LPI2C0_SCL	I2C Instance-0, Clock pin
11	PTE4	CAN0_RX	CAN0_RX	CAN Instance-0, Rx
12	PTE5	CAN0_TX	CAN0_TX	CAN Instance-0, Tx

The screenshot displays the S32 Design Studio IDE interface for the S32K144 microcontroller. The main window is divided into several panels:

- Pins Table:** A table listing pins and their functions. The 'Pins' tab is selected. The table has columns: Pin, Pin name, Label, Identifier, PORT, FTM, and LPSPI. Pins 21 and 22 are highlighted in green.
- Routing Details:** A table showing the routing for the BOARD_InitPins functional group. The 'Routing Details for BOARD...' tab is selected. The table has columns: #, Peripheral, Signal, Arrow, Routed pin/signal, Label, Identifier, Power group, Direction, Interrupt Status, Interrupt Configuration, Lock Register, Pull Enable, Pull Select, and Digit. The first row is highlighted in blue.
- Configuration - General Info:** A panel showing the configuration of the S32K144 microcontroller. It includes fields for Processor (S32K144), Part number (S32K144_LQFP100), Core (Cortex-M4), and SDK Version (s32sdk_s32k1xx_rtm_403).
- Configuration - HW Info:** A panel showing the hardware configuration. It includes a 'Pins' section with a green toggle switch and a 'Generated code' section with a blue toggle switch. The 'Pins' section is expanded, showing a list of pins and their functions.
- Functional groups:** A panel showing the functional groups. The 'BOARD_InitPins' group is selected.
- Other tools:** A panel showing other tools. The 'S32K144_LQFP100 - LQFP 100 package' is selected.

Give name in Label, Identifier and Initial value column as shown below

Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

hello_world_s32k144 Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

Pin type filter text

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPI
13	VREFL					
14	VSS_14					
15	PTB7			PTB7		
16	PTB6			PTB6		
17	PTE14			PTE14	FTM0_FLT1[...]	
18	PTE3			PTE3	FTM0_FLT0[...]	
19	PTD12			PTE12	FTM0_FLT3	
20	PTD17			PTD17	FTM0_FLT2	
21	PTD16	RGB_LED_GREEN	RGB_LED_GREEN	PTD16	FTM0_CH1	LPSPI0_SIN
22	PTD15	RGB_LED_RED	RGB_LED_RED	PTD15	FTM0_CH0	LPSPI0_SCK
23	PTD9			PTE9	FTM0_CH7	
24	PTD14			PTD14	FTM2_CH5	
25	PTD13			PTD13	FTM2_CH4	
26	PTE8			PTE8	FTM0_CH6	
27	PTB5			PTB5	FTM0_CH5	LPSPI0_PCS1[...]
28	PTB4			PTB4	FTM0_CH4	LPSPI0_SOUT
29	PTC3			PTC3	FTM0_CH3	
30	PTC2			PTC2	FTM0_CH2	
31	PTD7			PTD7	FTM2_FLT3	
32	PTD6			PTD6	FTM2_FLT2	

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 4

Peripheral	Signal	Arrow	Routed pin/sig...	Label	Identifier	Power...	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter	Drive Strength	Passive Filter	Initial Valu
PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	n/a	n/a	Low
PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	n/a	n/a	Low
PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	Low	n/a	High
PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	Low	n/a	High

Configuration - G

Configuration - H

Processor: S32K1

Part number: S32K1

Core: Cortex

SDK Version: s32sdl

Project

Pins

Config electric and ru

Generated code

Update code ena

board\pin_mux.

board\pin_mux.

Functional group

BOARD_InitPins

Other tools

Problems

type filter text

Level Resource

258M of 1017M

Click on Update Code

Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

hello_world_s32k144 Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

Pin Pin name Label Identifier POI

Pin	Pin name	Label	Identifier	POI
☒	13	VREFL		
☒	14	VSS_14		
☐	15	PTB7		PTE
☐	16	PTB6		PTE
☐	17	PTE14		PTE
☐	18	PTE3		PTE
☐	19	PTE12		PTE
☐	20	PTD17		PTC
☒	21	PTD16	RGB_LED_GREEN	PTC
☒	22	PTD15	RGB_LED_RED	PTC
☐	23	PTE9		PTE
☐	24	PTD14		PTC
☐	25	PTD13		PTC
☐	26	PTE8		PTE
☐	27	PTB5		PTE
☐	28	PTB4		PTE
☐	29	PTC3		PTC
☐	30	PTC2		PTC
☐	31	PTD7		PTC
☐	32	PTD6		PTC

S32K144_LQFP100 - LQFP 100 package

Routing Details

Pins Signals type filter text

Peripheral	Signal	Arrow	Routed pin/sig...	Label	Identifier	Power...	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter	Drive
PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	n/a
PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	n/a
PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	Low
PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	Low

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

☒ Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue
-------	----------	-------

hello_world_s32k144 368M of 1017M

Click on OK

Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

hello_world_s32k144

Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

Pin Pin name Label Identifier POI

Pin	Pin name	Label	Identifier	POI
☒	13	VREFL		
☒	14	VSS_14		
☐	15	PTB7		PTE
☐	16	PTB6		PTE
☐	17	PTE14		PTE
☐	18	PTE3		PTE
☐	19	PTE12		PTE
☐	20	PTD17		PTC
☒	21	PTD16	RGB_LED_GREEN	PTC
☒	22	PTD15	RGB_LED_RED	PTC
☐	23	PTE9		PTE
☐	24	PTD14		PTC
☐	25	PTD13		PTC
☐	26	PTE8		PTE
☐	27	PTB5		PTE
☐	28	PTB4		PTE
☐	29	PTC3		PTC
☐	30	PTC2		PTC
☐	31	PTD7		PTC
☐	32	PTD6		PTC

Update Files

Generated file	Status
☒ Pins	
☒ board\pin_mux.c	change
☒ board\pin_mux.h	change
☒ Clocks	
☒ board\clock_config.c	change
☒ board\clock_config.h	change
☒ Peripherals	
☒ board\sdk_project_config.h	no change
☐ DCD	warnings
☐ IVT	warnings
☐ QuadSPI	warnings
☐ DDR	warnings
☒ Configuration	
hello_world_s32k144.mex	change

Options

☒ Always show details before Update Code

OK Cancel

Routing Details for BOARD... 4

Peripheral	Signal	Arrow	Routed pin/sig...	Label	Identifier	Power	Direction	Interrupt status	Interrupt Configuration	Lock Register	Pin Enable	Pull Select	Digital Filter	Drive
PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	n/a
PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	n/a
PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	Low
PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	Low

hello_world_s32k144

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Overview Code Preview Registers

Configuration - General Info

Configuration - HW Info

Processor: S32K144
Part number: S32K144_LQFP100
Core: Cortex-M4
SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

☒ Update code enabled

board\pin_mux.c
board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue
-------	----------	-------

Click on C/C++

Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

hello_world_s32k144 Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

type filter text

Pin	Pin name	Label	Identifier	POI
☒	13	VREFL		
☒	14	VSS_14		
☐	15	PTB7		PTE
☐	16	PTB6		PTE
☐	17	PTE14		PTE
☐	18	PTE3		PTE
☐	19	PTE12		PTE
☐	20	PTD17		PTC
☒	21	PTD16	RGB_LED_GREEN	PTI
☒	22	PTD15	RGB_LED_RED	PTI
☐	23	PTE9		PTE
☐	24	PTD14		PTC
☐	25	PTD13		PTC
☐	26	PTE8		PTE
☐	27	PTB5		PTE
☐	28	PTB4		PTE
☐	29	PTC3		PTC
☐	30	PTC2		PTC
☐	31	PTD7		PTC
☐	32	PTD6		PTC

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 4

Peripheral	Signal	Arrow	Routed pin/sig...	Label	Identifier	Power...	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter	Drive
PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	n/a
PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	n/a
PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	Low
PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF...	Unlocked	Disabled	Pull Down	Disabled	Low

hello_world_s32k144

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Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

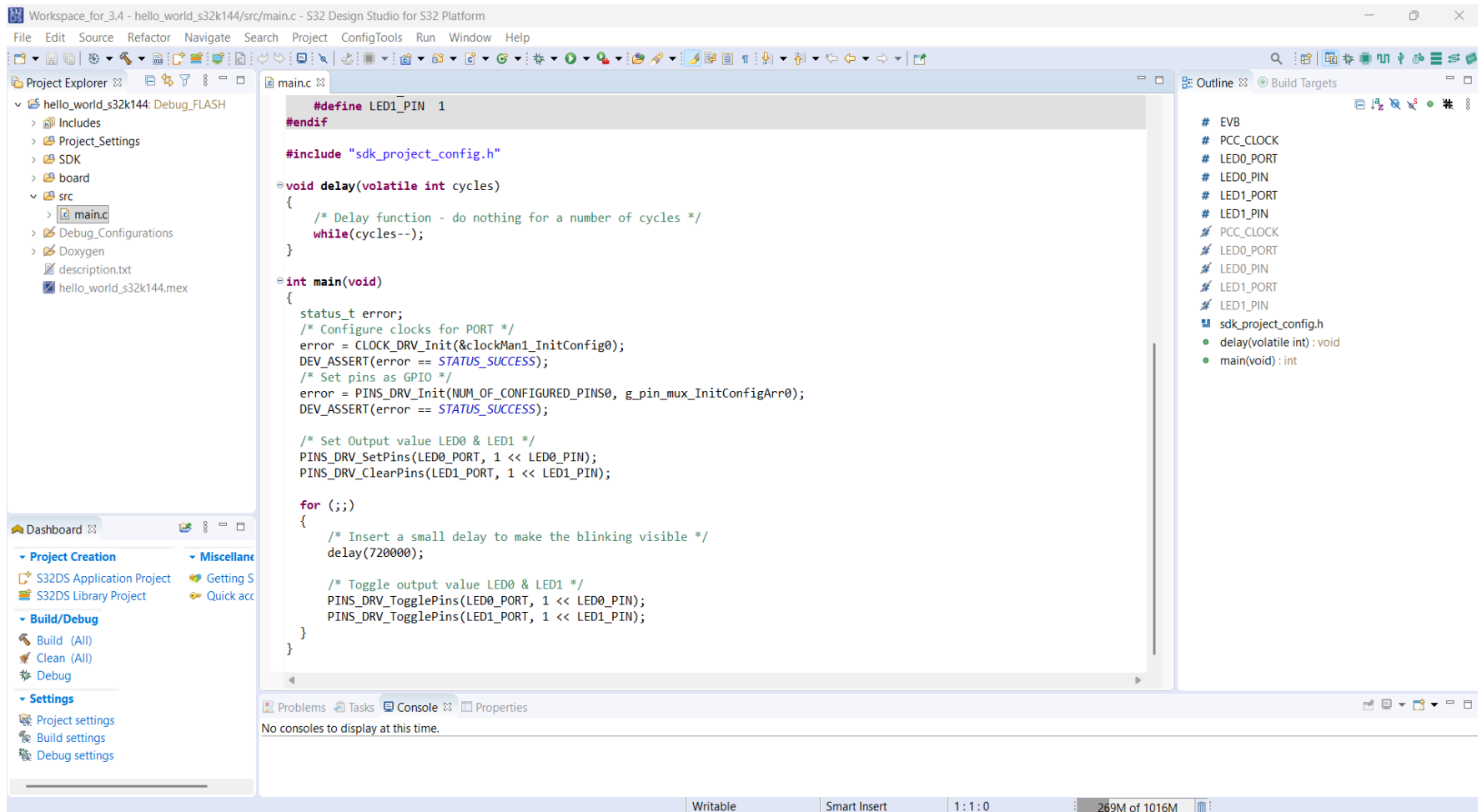
Other tools

Problems

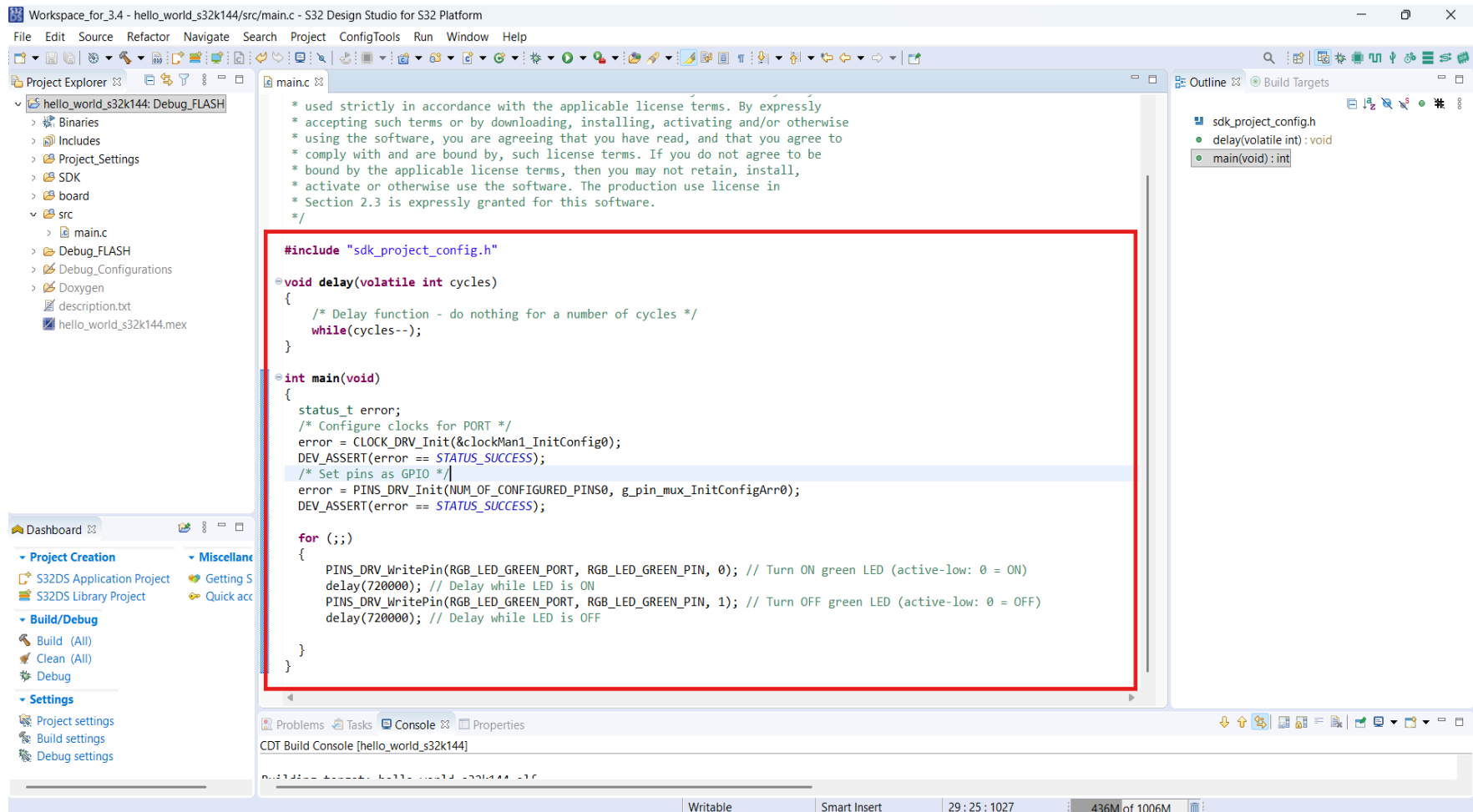
type filter text

Level	Resource	Issue
-------	----------	-------

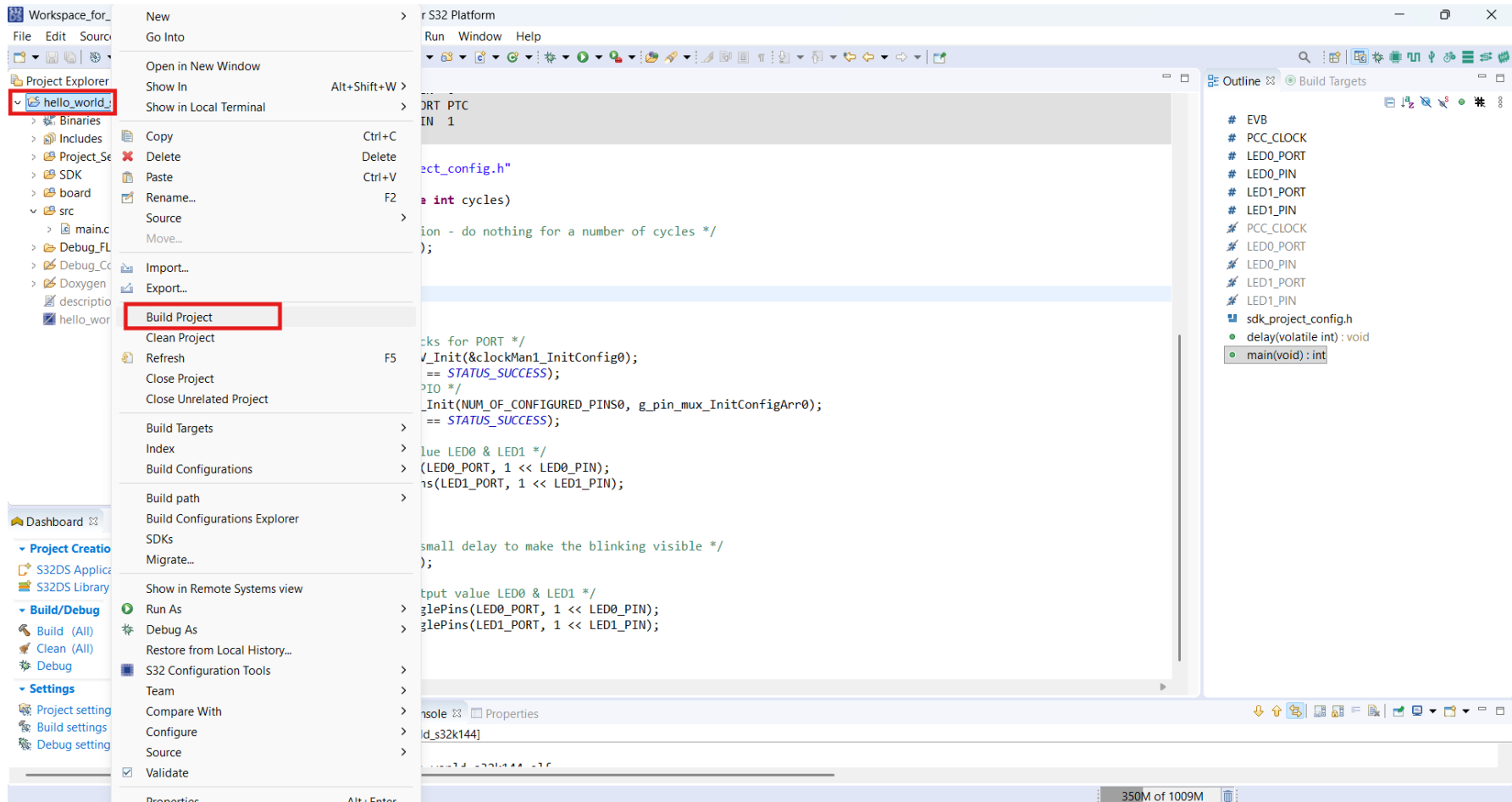
It will open C/C++ perspective



Write code and save project



Select project and right click and click on Build project



Check for any errors

The screenshot displays the S32 Design Studio IDE interface. The main editor shows the source code for `main.c` in the `hello_world_s32k144` project. The code includes a `delay` function and a `main` function that configures clocks, sets pins, and toggles an LED. The Project Explorer on the left shows the project structure. The Outline view on the right lists the functions in the project. The Problems view at the bottom is empty, indicating no errors. The Console view shows the build output, which includes the command to build the target and the resulting file sizes. The status bar at the bottom indicates the build is finished with 0 errors and 0 warnings.

```
void delay(volatile int cycles)
{
    /* Delay function - do nothing for a number of cycles */
    while(cycles--);
}

int main(void)
{
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);

    for (;;)
    {
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 0); // Turn ON green LED (active-low: 0 = ON)
        delay(720000); // Delay while LED is ON
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 1); // Turn OFF green LED (active-low: 0 = OFF)
        delay(720000); // Delay while LED is OFF
    }
}
```

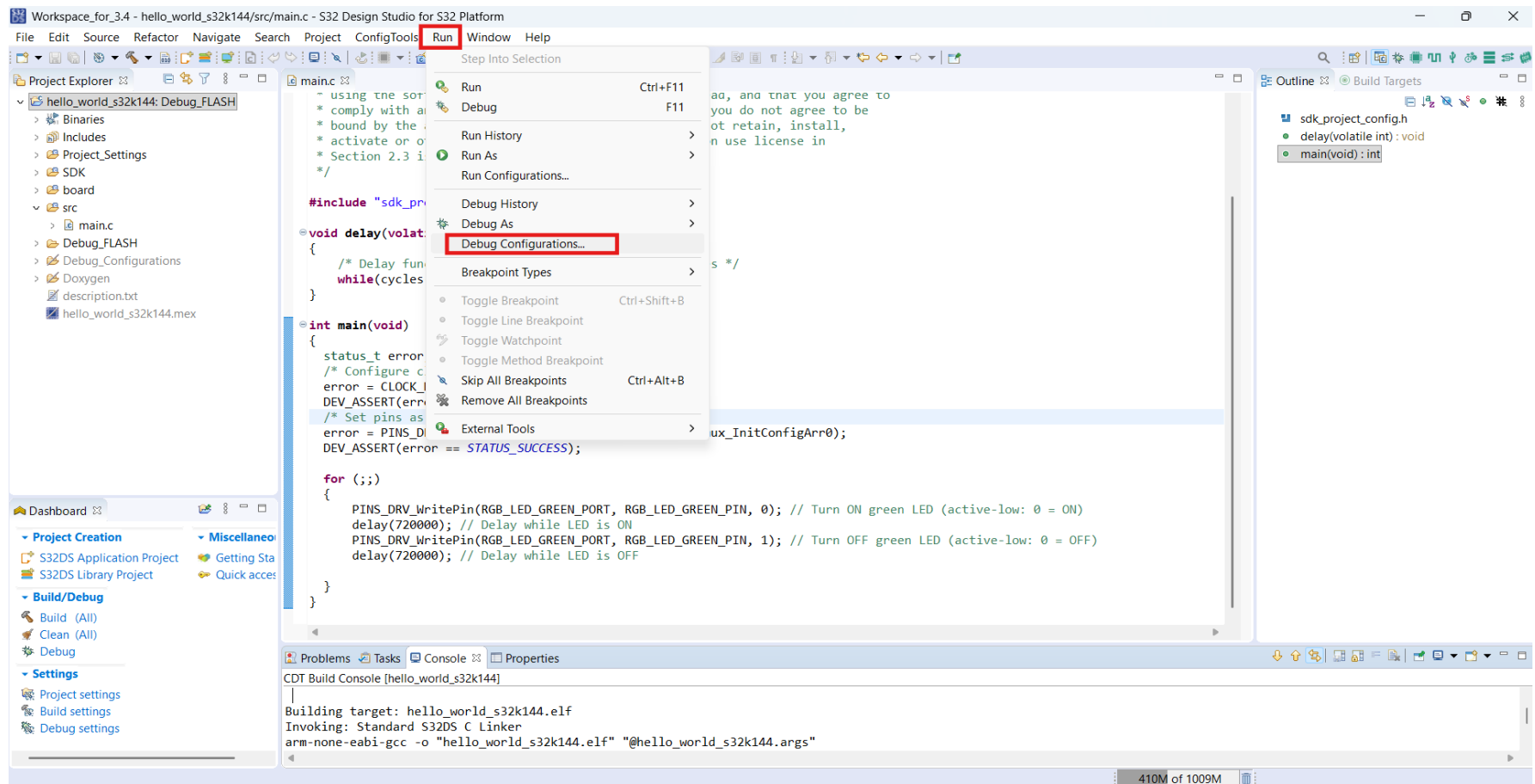
Building target: hello_world_s32k144.elf
Invoking: Standard S32DS C Linker
arm-none-eabi-gcc -o "hello_world_s32k144.elf" "@hello_world_s32k144.args"
Finished building target: hello_world_s32k144.elf

Invoking: Standard S32DS Print Size
arm-none-eabi-size --format=berkeley hello_world_s32k144.elf
text data bss dec hex filename
5776 552 3096 9424 24d0 hello_world_s32k144.elf
Finished building: hello_world_s32k144.siz

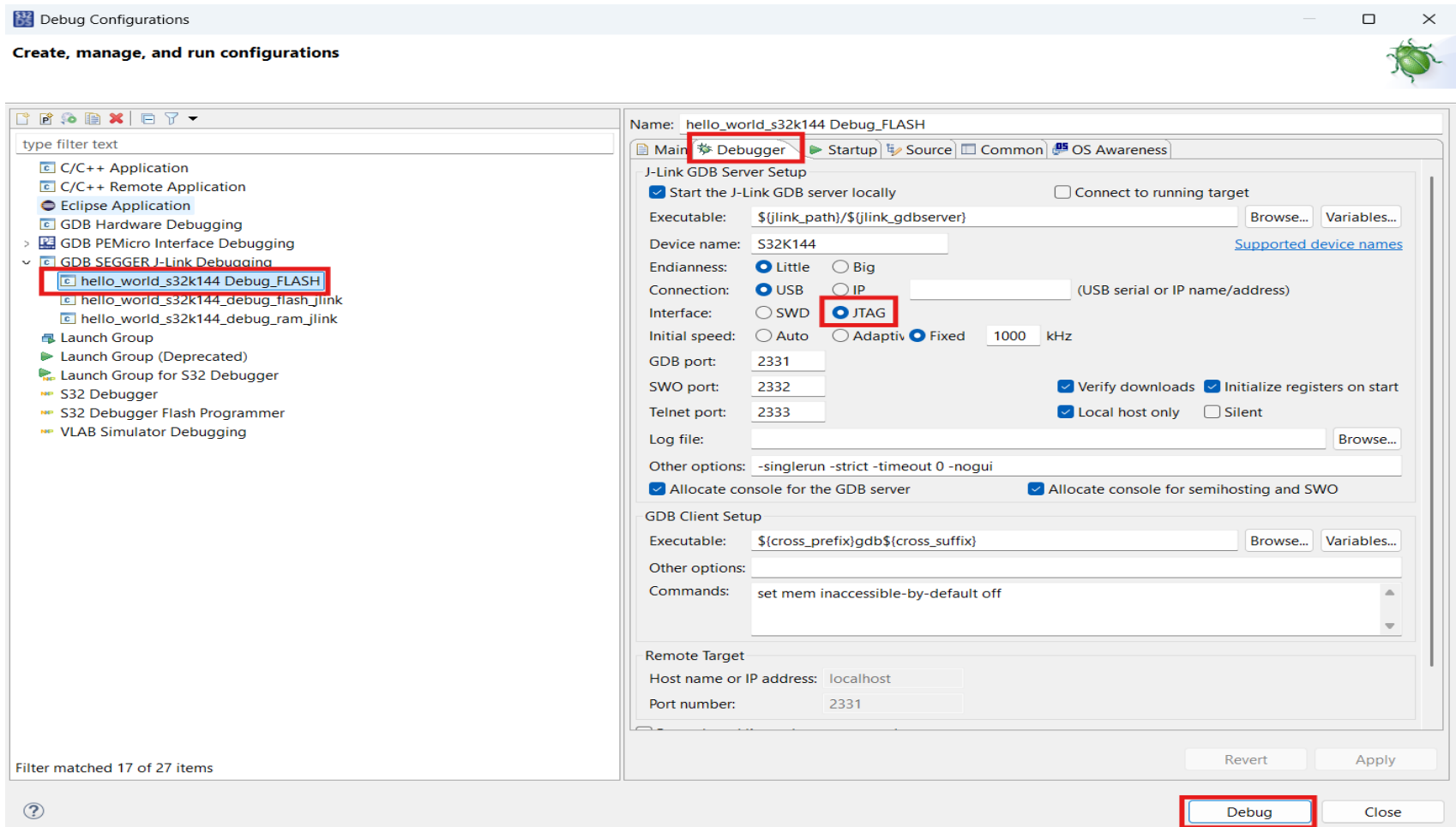
16:15:17 Build Finished. 0 errors, 0 warnings. (took 1s.457ms)

Connect S32K144 based development board to laptop/PC.

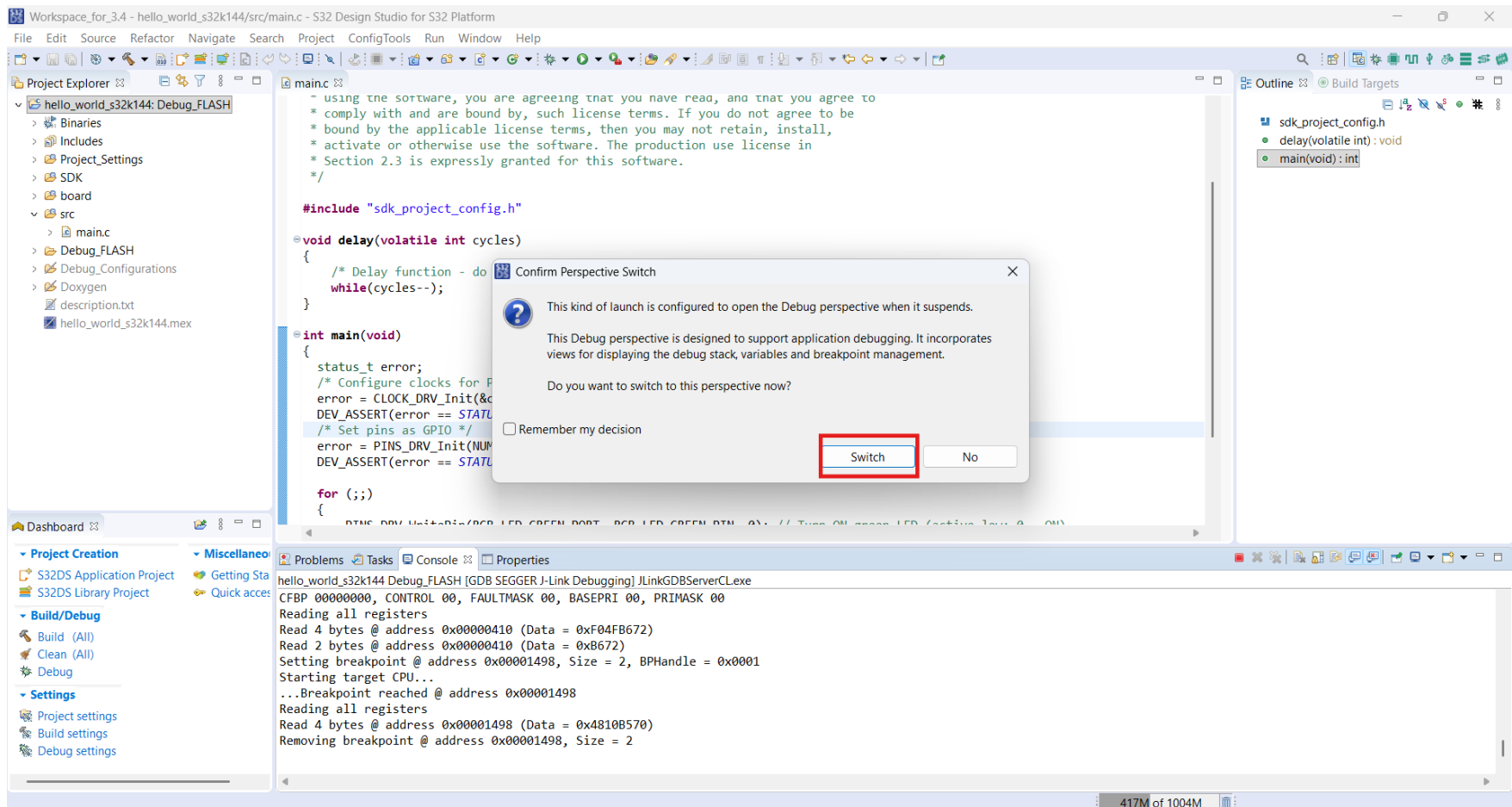
Flash using debugger: Run -> Debug Configuration



Click on Debugger and do settings as below and Click on Debug



Click on Switch



Open Debug perspective

Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Run FreeRTOS Window Help

Debug Project Explorer

- hello_world_s32k144 Debug_FLASH [GDB SEGGER J-L
- hello_world_s32k144.elf
 - Thread #1 57005 (Suspended : Breakpoint)
 - main() at main.c:24 0x1498
 - JLinkGDBServerCLex
 - arm-none-eabi-gdb
 - Semihosting and SWV

main.c

```
/* Section 2.3 is expressly granted for this software.
*/

#include "sdk_project_config.h"

void delay(volatile int cycles)
{
    /* Delay function - do nothing for a number of cycles */
    while(cycles--);
}

int main(void)
{
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);

    for (;;)
    {
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 0); // Turn ON green LED (active-low)
        delay(720000); // Delay while LED is ON
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 1); // Turn OFF green LED (active-low)
        delay(720000); // Delay while LED is OFF
    }
}
```

Expression Type Value

Add new expression

Outline

- sdk_project_config.h
- delay(volatile int): void
- main(void): int

Console Registers Problems Executables Debug Shell Watch registers Debugger Console Memory Spaces

hello_world_s32k144 Debug_FLASH [GDB SEGGER J-Link Debugging] Semihosting and SWV

SEGGER J-Link GDB Server V6.42a - Terminal output channel

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Click on Resume, starts debugging

The screenshot shows the S32 Design Studio for S32 Platform IDE. The main window displays the source code for `main.c`. A red arrow points to the **Resume** button (a green play icon) in the top toolbar. The left sidebar shows the **Project Explorer** with the project `hello_world_s32k144` and the `main()` function selected. The bottom status bar indicates the current location is `main.c:24 0x1498`.

```
/* Section 2.3 is expressly granted for this software.
 */

#include "sdk_project_config.h"

void delay(volatile int cycles)
{
    /* Delay function - do nothing for a number of cycles */
    while(cycles--);
}

int main(void)
{
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);

    for (;;)
    {
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 0); // Turn ON green LED (active-low)
        delay(720000); // Delay while LED is ON
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 1); // Turn OFF green LED (active-low)
        delay(720000); // Delay while LED is OFF
    }
}
```

The right sidebar shows the **Outline** view with the following structure:

- sdk_project_config.h
- delay(volatile int): void
- main(void): int

The bottom status bar shows the current location is `main.c:24 0x1498`.

Check the output on connected development board

Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Run FreeRTOS Window Help

Debug Project Explorer

hello_world_s32k144 Debug_FLASH [GDB SEGGER J-L]

hello_world_s32k144.elf

Thread #1 57005 (Running : User Request)

JLinkGDBServerCLex

arm-none-eabi-gdb

Semihosting and SWV

main.c

```
/* Section 2.3 is expressly granted for this software. */
#include "sdk_project_config.h"

void delay(volatile int cycles)
{
    /* Delay function - do nothing for a number of cycles */
    while(cycles--);
}

int main(void)
{
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);

    for (;;)
    {
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 0); // Turn ON green LED (active-low)
        delay(720000); // Delay while LED is ON
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 1); // Turn OFF green LED (active-low)
        delay(720000); // Delay while LED is OFF
    }
}
```

Expression Type Value

+ Add new expression

Outline

- sdk_project_config.h
- delay(volatile int): void
- main(void): int

Dashboard

- Project Creation
 - S32DS Application Project
 - S32DS Library Project
- Build/Debug
 - Build (All)
 - Clean (All)
 - Debug
- Settings
 - Project settings
 - Build settings
 - Debug settings
- Miscellaneous
 - Getting Started
 - Quick access

Console Registers Problems Executables Debug Shell Watch registers Debugger Console Memory Spaces

hello_world_s32k144 Debug_FLASH [GDB SEGGER J-Link Debugging] JLinkGDBServerCLex

Read 2 bytes @ address 0x00000410 (Data = 0xB672)

Setting breakpoint @ address 0x00001498, Size = 2, BPHandle = 0x0001

Starting target CPU...

...Breakpoint reached @ address 0x00001498

Reading all registers

Read 4 bytes @ address 0x00001498 (Data = 0x4810B570)

Removing breakpoint @ address 0x00001498, Size = 2

Starting target CPU...

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Click on Terminate to stop debugging

The screenshot shows the S32 Design Studio IDE interface. A red arrow points to the 'Terminate' button (a red square with a white 'X') in the top toolbar. The main editor displays the C source code for 'main.c'. The Project Explorer on the left shows the project structure. The Console window at the bottom shows the debug output.

```
/* Section 2.3 is expressly granted for this software. */  
#include "sdk_project_config.h"  
  
void delay(volatile int cycles)  
{  
    /* Delay function - do nothing for a number of cycles */  
    while(cycles--);  
}  
  
int main(void)  
{  
    status_t error;  
    /* Configure clocks for PORT */  
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);  
    DEV_ASSERT(error == STATUS_SUCCESS);  
    /* Set pins as GPIO */  
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);  
    DEV_ASSERT(error == STATUS_SUCCESS);  
  
    for (;;) {  
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 0); // Turn ON green LED (active-low)  
        delay(720000); // Delay while LED is ON  
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 1); // Turn OFF green LED (active-low)  
        delay(720000); // Delay while LED is OFF  
    }  
}
```

Console Output:

```
hello_world_s32k144 Debug_FLASH [GDB SEGGER J-Link Debugging] JLinkGDBServerCLExe  
Read 2 bytes @ address 0x00000410 (Data = 0xB672)  
Setting breakpoint @ address 0x00001498, Size = 2, BPHandle = 0x0001  
Starting target CPU...  
...Breakpoint reached @ address 0x00001498  
Reading all registers  
Read 4 bytes @ address 0x00001498 (Data = 0x4810B570)  
Removing breakpoint @ address 0x00001498, Size = 2  
Starting target CPU...
```

Click on C/C++

The screenshot displays the S32 Design Studio IDE interface. The top toolbar features a red arrow pointing to the C/C++ icon. The main editor window shows the source code for `main.c` in the `hello_world_s32k144` project. The code includes a delay function and a `main` function that initializes the clock and pins, and then turns on and off a green LED. The left sidebar shows the Project Explorer with the project structure. The bottom status bar indicates the memory usage: 318M of 1007M.

```
Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform
File Edit Source Refactor Navigate Search Project ConfigTools Run FreeRTOS Window Help

Debug Project Explorer

hello_world_s32k144 Debug_FLASH [GDB SEGGER J-L
  hello_world_s32k144.elf
    Thread #1 57005 (Running : User Request)
    JLinkGDBServerCLExe
    arm-none-eabi-gdb
    Semihosting and SWV

main.c
  * Section 2.3 is expressly granted for this software.
  */
  #include "sdk_project_config.h"
  void delay(volatile int cycles)
  {
    /* Delay function - do nothing for a number of cycles */
    while(cycles--);
  }
  int main(void)
  {
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);

    for (;;)
    {
      PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 0); // Turn ON green LED (active-low)
      delay(720000); // Delay while LED is ON
      PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 1); // Turn OFF green LED (active-low)
      delay(720000); // Delay while LED is OFF
    }
  }

Dashboard
  Project Creation
    S32DS Application Project
    S32DS Library Project
  Build/Debug
    Build (All)
    Clean (All)
    Debug
  Settings
    Project settings
    Build settings
    Debug settings
  Miscellaneous
    Getting Started
    Quick access

Console Registers Problems Executables Debug Shell Watch registers Debugger Console Memory Spaces
hello_world_s32k144 Debug_FLASH [GDB SEGGER J-Link Debugging] JLinkGDBServerCLExe
Read 2 bytes @ address 0x00000410 (Data = 0xB672)
Setting breakpoint @ address 0x00001498, Size = 2, BPHandle = 0x0001
Starting target CPU...
...Breakpoint reached @ address 0x00001498
Reading all registers
Read 4 bytes @ address 0x00001498 (Data = 0x4810B570)
Removing breakpoint @ address 0x00001498, Size = 2
Starting target CPU...

318M of 1007M
```

The screenshot displays the S32 Design Studio IDE interface. The main editor window shows the source code for `main.c` in the `hello_world_s32k144: Debug_FLASH` project. The code includes a license header, a header file `sdk_project_config.h`, and two functions: `delay` and `main`.

```
#include "sdk_project_config.h"

void delay(volatile int cycles)
{
    /* Delay function - do nothing for a number of cycles */
    while(cycles--);
}

int main(void)
{
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);

    for (;;)
    {
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 0); // Turn ON green LED (active-low: 0 = ON)
        delay(720000); // Delay while LED is ON
        PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 1); // Turn OFF green LED (active-low: 0 = OFF)
        delay(720000); // Delay while LED is OFF
    }
}
```

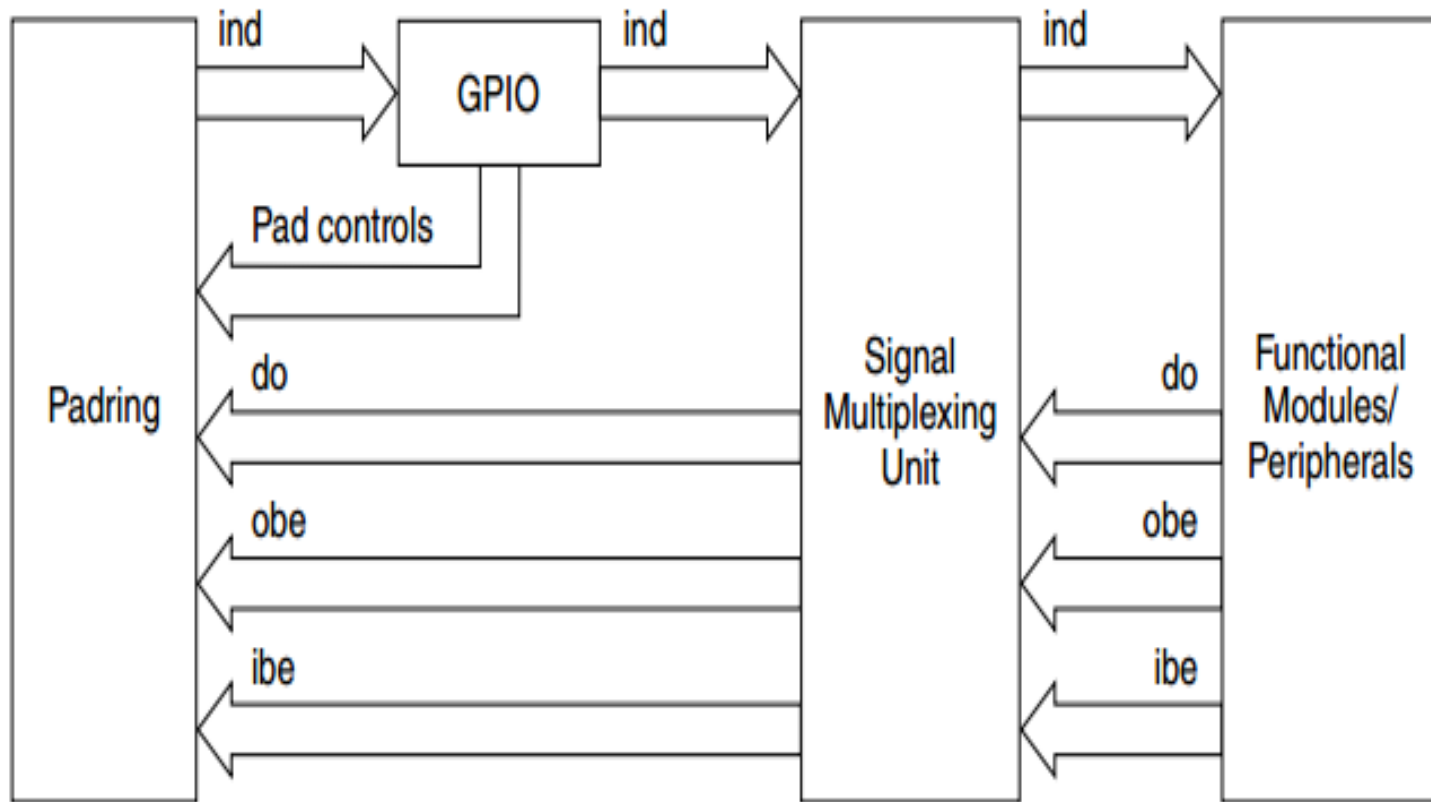
The Project Explorer on the left shows the project structure, including `Binaries`, `Includes`, `Project_Settings`, `SDK`, `board`, `src`, `main.c`, `Debug_FLASH`, `Debug_Configurations`, `Doxygen`, `description.txt`, and `hello_world_s32k144.mex`.

The Outline on the right shows the project structure, including `sdk_project_config.h`, `delay(volatile int):void`, and `main(void):int`.

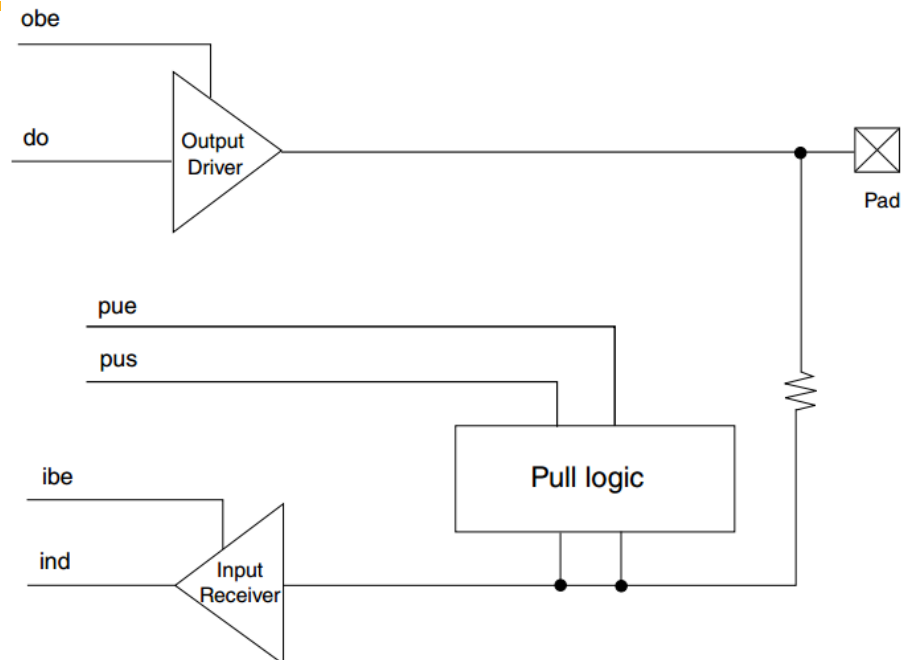
The Console at the bottom shows the output of the program, indicating that the connection was closed by the GDB server.

```
<terminated> hello_world_s32k144 Debug_FLASH [GDB SEGGER J-Link Debugging] Semihosting and SWV
SEGGER J-Link GDB Server V6.42a - Terminal output channel
Connection closed by the GDB server.
```


Signal Multiplexing



GPIO pad representation



Signal name	Direction	Description
pad	I/O	I/O to external world
do	I	Data coming from the core into the pad
obe	I	Enable output driver
pue	I	0: Disable internal pullup or pulldown resistor 1: Enable internal pullup or pulldown resistor
pus	I	0: Enable internal pulldown resistor if pue is set 1: Enable internal pullup resistor if pue is set
ibe	I	Enable input receiver
ind	O	Data coming out of the pad into the core

Kind Courtesy of reference books/papers/Links:

1. <https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-s32-platform:S32DS-S32PLATFORM>

Thank you